Solution Since the input sequence $x(n)$ is a constant for $n \geq 0$, the form of the solution that we assume is also a constant. Hence the assumed solution of the difference equation to the forcing function $x(n)$, called the *particular solution* of the difference equation, is

$$y_p(n) = Ku(n)$$

where K is a scale factor determined so that (2.4.26) is satisfied. Upon substitution of this assumed solution into (2.4.26), we obtain

$$Ku(n) + a_1 Ku(n-1) = u(n)$$

To determine K, we must evaluate this equation for any $n \geq 1$, where none of the terms vanish. Thus

$$K + a_1 K = 1$$

$$K = \frac{1}{1+a_1}$$

Therefore, the particular solution to the difference equation is

$$y_p(n) = \frac{1}{1+a_1} u(n) \tag{2.4.27}$$

In this example, the input $x(n)$, $n \geq 0$, is a constant and the form assumed for the particular solution is also a constant. If $x(n)$ is an exponential, we would assume that the particular solution is also an exponential. If $x(n)$ were a sinusoid, then $y_p(n)$ would also be a sinusoid. Thus our assumed form for the particular solution takes the basic form of the signal $x(n)$. Table 2.1 provides the general form of the particular solution for several types of excitation.

Example 2.4.7

Determine the particular solution of the difference equation

$$y(n) = \tfrac{5}{6}y(n-1) - \tfrac{1}{6}y(n-2) + x(n)$$

when the forcing function $x(n) = 2^n$, $n \geq 0$ and zero elsewhere.

TABLE 2.1 GENERAL FORM OF THE PARTICULAR SOLUTION FOR SEVERAL TYPES OF INPUT SIGNALS

Input Signal, $x(n)$	Particular Solution, $y_p(n)$
A (constant)	K
AM^n	KM^n
An^M	$K_0 n^M + K_1 n^{M-1} + \cdots + K_M$
$A^n n^M$	$A^n(K_0 n^M + K_1 n^{M-1} + \cdots + K_M)$
$\left\{\begin{matrix} A\cos\omega_0 n \\ A\sin\omega_0 n \end{matrix}\right\}$	$K_1\cos\omega_0 n + K_2\sin\omega_0 n$

Solution The form of the particular solution is

$$y_p(n) = K2^n \qquad n \geq 0$$

Upon substitution of $y_p(n)$ into the difference equation, we obtain

$$K2^n u(n) = \tfrac{5}{6}K2^{n-1}u(n-1) - \tfrac{1}{6}K2^{n-2}u(n-2) + 2^n u(n)$$

To determine the value of K, we can evaluate this equation for any $n \geq 2$, where none of the terms vanish. Thus we obtain

$$4K = \tfrac{5}{6}(2K) - \tfrac{1}{6}K + 4$$

and hence $K = \frac{8}{5}$. Therefore, the particular solution is

$$y_p(n) = \tfrac{8}{5}2^n \qquad n \geq 0$$

We have now demonstrated how to determine the two components of the solution to a difference equation with constant coefficients. These two components are the homogeneous solution and the particular solution. From these two components, we construct the total solution from which we can obtain the zero-state response.

The total solution of the difference equation. The linearity property of the linear constant-coefficient difference equation allows us to add the homogeneous solution and the particular solution in order to obtain the *total solution*. Thus

$$y(n) = y_h(n) + y_p(n)$$

The resultant sum $y(n)$ contains the constant parameters $\{C_i\}$ embodied in the homogeneous solution component $y_h(n)$. These constants can be determined to satisfy the initial conditions. The following example illustrates the procedure.

Example 2.4.8

Determine the total solution $y(n)$, $n \geq 0$, to the difference equation

$$y(n) + a_1 y(n-1) = x(n) \tag{2.4.28}$$

when $x(n)$ is a unit step sequence [i.e., $x(n) = u(n)$] and $y(-1)$ is the initial condition.

Solution From (2.4.19) of Example 2.4.4, the homogeneous solution is

$$y_h(n) = C(-a_1)^n$$

and from (2.4.26) of Example 2.4.6, the particular solution is

$$y_p(n) = \frac{1}{1 + a_1}$$

Consequently, the total solution is

$$y(n) = C(-a_1)^n + \frac{1}{1 + a_1} \qquad n \geq 0 \tag{2.4.29}$$

where the constant C is determined to satisfy the initial condition $y(-1)$.

In particular, suppose that we wish to obtain the zero-state response of the system described by the first-order difference equation in (2.4.28). Then we set $y(-1) = 0$. To evaluate C, we evaluate (2.4.28) at $n = 0$ obtaining

$$\begin{aligned} y(0) + a_1 y(-1) &= 1 \\ y(0) &= 1 \end{aligned}$$

On the other hand, (2.4.29) evaluated at $n = 0$ yields

$$y(0) = C + \frac{1}{1 + a_1}$$

Consequently,

$$\begin{aligned} C + \frac{1}{1 + a_1} &= 1 \\ C &= \frac{a_1}{1 + a_1} \end{aligned}$$

Substitution for C into (2.4.29) yields the zero-state response of the system

$$y_{zs}(n) = \frac{1 - (-a_1)^{n+1}}{1 + a_1} \qquad n \geq 0$$

If we evaluate the parameter C in (2.4.29) under the condition that $y(-1) \neq 0$, the total solution will include the zero-input response as well as the zero-state response of the system. In this case (2.4.28) yields

$$\begin{aligned} y(0) + a_1 y(-1) &= 1 \\ y(0) &= -a_1 y(-1) + 1 \end{aligned}$$

On the other hand, (2.4.29) yields

$$y(0) = C + \frac{1}{1 + a_1}$$

By equating these two relations, we obtain

$$\begin{aligned} C + \frac{1}{1 + a_1} &= -a_1 y(-1) + 1 \\ C &= -a_1 y(-1) + \frac{a_1}{1 + a_1} \end{aligned}$$

Finally, if we substitute this value of C into (2.4.29), we obtain

$$\begin{aligned} y(n) &= (-a_1)^{n+1} y(-1) + \frac{1 - (-a_1)^{n+1}}{1 + a_1} \qquad n \geq 0 \\ &= y_{zi}(n) + y_{zs}(n) \end{aligned} \tag{2.4.30}$$

We observe that the system response as given by (2.4.30) is consistent with the response $y(n)$ given in (2.4.8) for the first-order system (with $a = -a_1$), which was obtained by solving the difference equation iteratively. Furthermore, we note that the value of the constant C depends both on the initial condition $y(-1)$ and on the excitation function. Consequently, the value of C influences both the zero-input response and the zero-state response. On the other hand, if we wish to obtain the zero-state response only, we simply solve for C under the condition that $y(-1) = 0$, as demonstrated in Example 2.4.8.

We further observe that the particular solution to the difference equation can be obtained from the zero-state response of the system. Indeed, if $|a_1| < 1$, which is the condition for stability of the system, as will be shown in Section 2.4.4, the limiting value of $y_{zs}(n)$ as n approaches infinity, is the particular solution, that is,

$$y_p(n) = \lim_{n\to\infty} y_{zs}(n) = \frac{1}{1+a_1}$$

Since this component of the system response does not go to zero as n approaches infinity, it is usually called the *steady-state response* of the system. This response persists as long as the input persists. The component that dies out as n approaches infinity is called the *transient response* of the system.

Example 2.4.9

Determine the response $y(n)$, $n \geq 0$, of the system described by the second-order difference equation

$$y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1) \tag{2.4.31}$$

when the input sequence is

$$x(n) = 4^n u(n)$$

Solution We have already determined the solution to the homogeneous difference equation for this system in Example 2.4.5. From (2.4.22) we have

$$y_h(n) = C_1(-1)^n + C_2(4)^n \tag{2.4.32}$$

The particular solution to (2.4.31) is assumed to be an exponential sequence of the same form as $x(n)$. Normally, we could assume a solution of the form

$$y_p(n) = K(4)^n u(n)$$

However, we observe that $y_p(n)$ is already contained in the homogeneous solution, so that this particular solution is redundant. Instead, we select the particular solution to be linearly independent of the terms contained in the homogeneous solution. In fact, we treat this situation in the same manner as we have already treated multiple roots in the characteristic equation. Thus we assume that

$$y_p(n) = Kn(4)^n u(n) \tag{2.4.33}$$

Upon substitution of (2.4.33) into (2.4.31), we obtain

$$Kn(4)^n u(n) - 3K(n-1)(4)^{n-1}u(n-1) - 4K(n-2)(4)^{n-2}u(n-2)$$
$$= (4)^n u(n) + 2(4)^{n-1}u(n-1)$$

To determine K, we evaluate this equation for any $n \geq 2$, where none of the unit step terms vanish. To simplify the arithmetic, we select $n = 2$, from which we obtain $K = \frac{6}{5}$. Therefore,

$$y_p(n) = \tfrac{6}{5}n(4)^n u(n) \tag{2.4.34}$$

The total solution to the difference equation is obtained by adding (2.4.32) to (2.4.34). Thus

$$y(n) = C_1(-1)^n + C_2(4)^n + \tfrac{6}{5}n(4)^n \qquad n \geq 0 \tag{2.4.35}$$

where the constants C_1 and C_2 are determined such that the initial conditions are satisfied. To accomplish this, we return to (2.4.31), from which we obtain

$$\begin{aligned} y(0) &= 3y(-1) + 4y(-2) + 1 \\ y(1) &= 3y(0) + 4y(-1) + 6 \\ &= 13y(-1) + 12y(-2) + 9 \end{aligned}$$

On the other hand, (2.4.35) evaluated at $n = 0$ and $n = 1$ yields

$$\begin{aligned} y(0) &= C_1 + C_2 \\ y(1) &= -C_1 + 4C_2 + \tfrac{24}{5} \end{aligned}$$

We can now equate these two sets of relations to obtain C_1 and C_2. In so doing, we have the response due to initial conditions $y(-1)$ and $y(-2)$ (the zero-input response), and the zero-state or forced response.

Since we have already solved for the zero-input response in Example 2.4.5, we can simplify the computations above by setting $y(-1) = y(-2) = 0$. Then we have

$$\begin{aligned} C_1 + C_2 &= 1 \\ -C_1 + 4C_2 + \tfrac{24}{5} &= 9 \end{aligned}$$

Hence $C_1 = -\frac{1}{25}$ and $C_2 = \frac{26}{25}$. Finally, we have the zero-state response to the forcing function $x(n) = (4)^n u(n)$ in the form

$$y_{zs}(n) = -\tfrac{1}{25}(-1)^n + \tfrac{26}{25}(4)^n + \tfrac{6}{5}n(4)^n \qquad n \geq 0 \tag{2.4.36}$$

The total response of the system, which includes the response to arbitrary initial conditions, is the sum of (2.4.23) and (2.4.36).

2.4.4 The Impulse Response of a Linear Time-Invariant Recursive System

The impulse response of a linear time-invariant system was previously defined as the response of the system to a unit sample excitation [i.e., $x(n) = \delta(n)$]. In the case of a recursive system, $h(n)$ is simply equal to the zero-state response of the system when the input $x(n) = \delta(n)$ and the system is initially relaxed.

For example, in the simple first-order recursive system given in (2.4.7), the zero-state response given in (2.4.8), is

$$y_{zs}(n) = \sum_{k=0}^{n} a^k x(n-k) \tag{2.4.37}$$

With $x(n) = \delta(n)$ is substituted into (2.4.37), we obtain

$$\begin{aligned} y_{zs}(n) &= \sum_{k=0}^{n} a^k \delta(n-k) \\ &= a^n \qquad n \geq 0 \end{aligned}$$

Hence the impulse response of the first-order recursive system described by (2.4.7) is

$$h(n) = a^n u(n) \tag{2.4.38}$$

as indicated in Section 2.4.2.

In the general case of an arbitrary, linear time-invariant recursive system, the zero-state response expressed in terms of the convolution summation is

$$y_{zs}(n) = \sum_{k=0}^{n} h(k)x(n-k) \qquad n \geq 0 \tag{2.4.39}$$

When the input is an impulse [i.e., $x(n) = \delta(n)$], (2.4.39) reduces to

$$y_{zs}(n) = h(n) \tag{2.4.40}$$

Now, let us consider the problem of determining the impulse response $h(n)$ given a linear constant-coefficient difference equation description of the system. In terms of our discussion in the preceding subsection, we have established the fact that the total response of the system to any excitation function consists of the sum of two solutions of the difference equation: the solution to the homogeneous equation plus the particular solution to the excitation function. In the case where the excitation is an impulse, the particular solution is zero, since $x(n) = 0$ for $n > 0$, that is,

$$y_p(n) = 0$$

Consequently, the response of the system to an impulse consists only of the solution to the homogeneous equation, with the $\{C_k\}$ parameters evaluated to satisfy the initial conditions dictated by the impulse. The following example illustrates the procedure for obtaining $h(n)$ given the difference equation for the system.

Example 2.4.10

Determine the impulse response $h(n)$ for the system described by the second-order difference equation

$$y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1) \tag{2.4.41}$$

Solution We have already determined in Example 2.4.5 that the solution to the homogeneous difference equation for this system is

$$y_h(n) = C_1(-1)^n + C_2(4)^n \qquad n \geq 0 \tag{2.4.42}$$

Since the particular solution is zero when $x(n) = \delta(n)$, the impulse response of the system is simply given by (2.4.42), where C_1 and C_2 must be evaluated to satisfy (2.4.41).

For $n = 0$ and $n = 1$, (2.4.41) yields

$$\begin{aligned} y(0) &= 1 \\ y(1) &= 3y(0) + 2 = 5 \end{aligned}$$

where we have imposed the conditions $y(-1) = y(-2) = 0$, since the system must be relaxed. On the other hand, (2.4.42) evaluated at $n = 0$ and $n = 1$ yields

$$\begin{aligned} y(0) &= C_1 + C_2 \\ y(1) &= -C_1 + 4C_2 \end{aligned}$$

By solving these two sets of equations for C_1 and C_2, we obtain

$$C_1 = -\tfrac{1}{5}, \qquad C_2 = \tfrac{6}{5}$$

Therefore, the impulse response of the system is

$$h(n) = [-\tfrac{1}{5}(-1)^n + \tfrac{6}{5}(4)^n]u(n)$$

We make the observation that both the simple first-order recursive system and the second-order recursive system have impulse responses that are infinite in duration. In other words, both of these recursive systems are IIR systems. In fact, due to the recursive nature of the system, any recursive system described by a linear constant-coefficient difference equation is an IIR system. The converse is not true, however. That is, not every linear time-invariant IIR system can be described by a linear constant-coefficient difference equation. In other words, recursive systems described by linear constant-coefficient difference equations are a subclass of linear time-invariant IIR systems.

The extension of the approach that we have demonstrated for determining the impulse response of the first- and second-order systems. generalizes in a straightforward manner. When the system is described by an Nth-order linear difference equation of the type given in (2.4.13), the solution of the homogeneous equation is

$$y_h(n) = \sum_{k=1}^{N} C_k \lambda_k^n$$

when the roots $\{\lambda_k\}$ of the characteristic polynomial are distinct. Hence the impulse response of the system is identical in form, that is,

$$h(n) = \sum_{k=1}^{N} C_k \lambda_k^n \tag{2.4.43}$$

where the parameters $\{C_k\}$ are determined by setting the initial conditions $y(-1) = \cdots = y(-N) = 0$.

This form of $h(n)$ allows us to easily relate the stability of a system, described by an Nth-order difference equation, to the values of the roots of the characteristic polynomial. Indeed, since BIBO stability requires that the impulse response be absolutely summable, then, for a causal system, we have

$$\sum_{n=0}^{\infty} |h(n)| = \sum_{n=0}^{\infty} \left| \sum_{k=1}^{N} C_k \lambda_k^n \right| \le \sum_{k=1}^{N} |C_k| \sum_{n=0}^{\infty} |\lambda_k|^n$$

Now if $|\lambda_k| < 1$ for all k, then

$$\sum_{n=0}^{\infty} |\lambda_k|^n < \infty$$

and hence

$$\sum_{n=0}^{\infty} |h(n)| < \infty$$

On the other hand, if one or more of the $|\lambda_k| \geq 1$, $h(n)$ is no longer absolutely summable, and consequently, the system is unstable. Therefore, a necessary and sufficient condition for the stability of a causal IIR system described by a linear constant-coefficient difference equation, is that all roots of the characteristic polynomial be less than unity in magnitude. The reader may verify that this condition carries over to the case where the system has roots of multiplicity m.

2.5 IMPLEMENTATION OF DISCRETE-TIME SYSTEMS

Our treatment of discrete-time systems has been focused on the time-domain characterization and analysis of linear time-invariant systems described by constant-coefficient linear difference equations. Additional analytical methods are developed in the next two chapters, where we characterize and analyze LTI systems in the frequency domain. Two other important topics that will be treated later are the design and implementation of these systems.

In practice, system design and implementation are usually treated jointly rather than separately. Often, the system design is driven by the method of implementation and by implementation constraints, such as cost, hardware limitations, size limitations, and power requirements. At this point, we have not as yet developed the necessary analysis and design tools to treat such complex issues. However, we have developed sufficient background to consider some basic implementation methods for realizations of LTI systems described by linear constant-coefficient difference equations.

2.5.1 Structures for the Realization of Linear Time-Invariant Systems

In this subsection we describe structures for the realization of systems described by linear constant-coefficient difference equations. Additional structures for these systems are introduced in Chapter 7.

As a beginning, let us consider the first-order system

$$y(n) = -a_1 y(n-1) + b_0 x(n) + b_1 x(n-1) \tag{2.5.1}$$

which is realized as in Fig. 2.32a. This realization uses separate delays (memory) for both the input and output signal samples and it is called a *direct form I structure*. Note that this system can be viewed as two linear time-invariant systems in cascade. The first is a nonrecursive, system described by the equation

$$v(n) = b_0 x(n) + b_1 x(n-1) \tag{2.5.2}$$

whereas the second is a recursive system described by the equation

$$y(n) = -a_1 y(n-1) + v(n) \tag{2.5.3}$$

However, as we have seen in Section 2.3.4, if we interchange the order of the cascaded linear time-invariant systems, the overall system response remains the same. Thus if we interchange the order of the recursive and nonrecursive systems, we obtain an alternative structure for the realization of the system described by (2.5.1). The resulting system is shown in Fig. 2.32b. From this figure we obtain the two difference equations

$$w(n) = -a_1 w(n-1) + x(n) \tag{2.5.4}$$

$$y(n) = b_0 w(n) + b_1 w(n-1) \tag{2.5.5}$$

which provide an alternative algorithm for computing the output of the system described by the single difference equation given in (2.5.1). In other words, the two difference equations (2.5.4) and (2.5.5) are equivalent to the single difference equation (2.5.1).

A close observation of Fig. 2.32 reveals that the two delay elements contain the same input $w(n)$ and hence the same output $w(n-1)$. Consequently, these two elements can be merged into one delay, as shown in Fig. 2.32c. In contrast

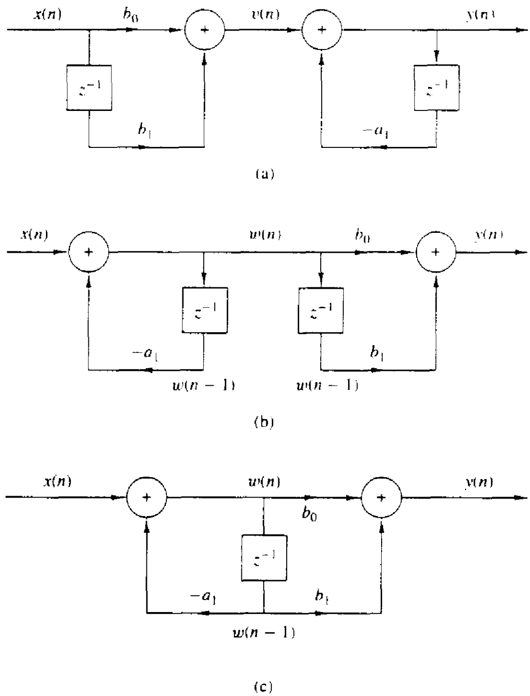


Figure 2.32 Steps in converting from the direct form I realization in (a) to the direct form II realization in (c).

to the direct form I structure, this new realization requires only one delay for the auxiliary quantity $w(n)$, and hence it is more efficient in terms of memory requirements. It is called the *direct form II structure* and it is used extensively in practical applications.

These structures can readily be generalized for the general linear time-invariant recursive system described by the difference equation

$$y(n) = -\sum_{k=1}^{N} a_k y(n-k) + \sum_{k=0}^{M} b_k x(n-k) \tag{2.5.6}$$

Figure 2.33 illustrates the direct form I structure for this system. This structure requires $M + N$ delays and $N + M + 1$ multiplications. It can be viewed as the cascade of a nonrecursive system

$$v(n) = \sum_{k=0}^{M} b_k x(n-k) \tag{2.5.7}$$

and a recursive system

$$y(n) = -\sum_{k=1}^{N} a_k y(n-k) + v(n) \tag{2.5.8}$$

By reversing the order of these two systems as was previously done for the first-order system, we obtain the direct form II structure shown in Fig. 2.34 for

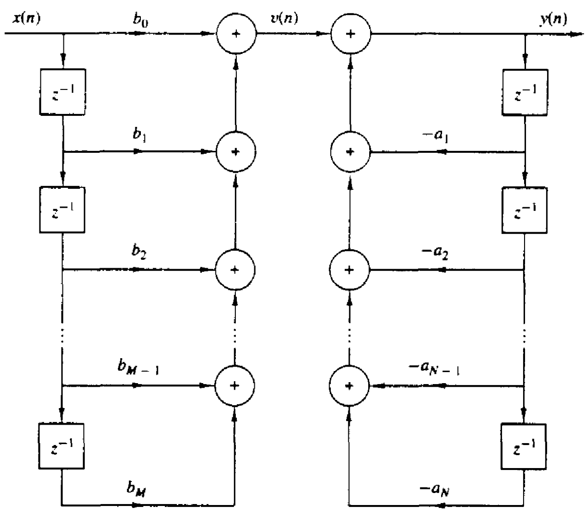


Figure 2.33 Direct form I structure of the system described by (2.5.6).

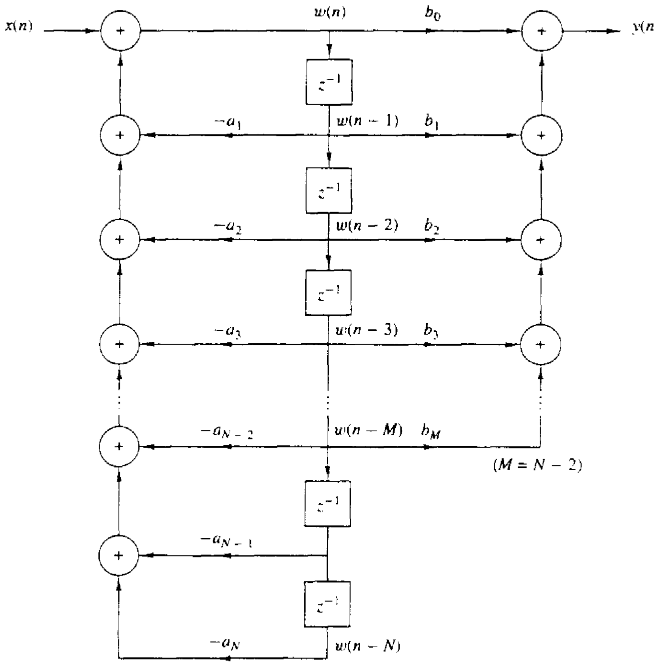


Figure 2.34 Direct form II structure for the system described by (2.5.6).

$N > M$. This structure is the cascade of a recursive system

$$w(n) = -\sum_{k=1}^{N} a_k w(n-k) + x(n) \tag{2.5.9}$$

followed by a nonrecursive system

$$y(n) = \sum_{k=0}^{M} b_k w(n-k) \tag{2.5.10}$$

We observe that if $N \geq M$, this structure requires a number of delays equal to the order N of the system. However, if $M > N$, the required memory is specified by M. Figure 2.34 can easily by modified to handle this case. Thus the direct form II structure requires $M + N + 1$ multiplications and $\max\{M, N\}$ delays. Because it requires the minimum number of delays for the realization of the system described by (2.5.6), it is sometimes called a *canonic form*.

A special case of (2.5.6) occurs if we set the system parameters $a_k = 0$, $k = 1, \ldots, N$. Then the input–output relationship for the system reduces to

$$y(n) = \sum_{k=0}^{M} b_k x(n-k) \tag{2.5.11}$$

which is a nonrecursive linear time-invariant system. This system views only the most recent $M+1$ input signal samples and, prior to addition, weights each sample by the appropriate coefficient b_k from the set $\{b_k\}$. In other words, the system output is basically a *weighted moving average* of the input signal. For this reason it is sometimes called a *moving average (MA) system*. Such a system is an FIR system with an impulse response h(k) equal to the coefficients b_k, that is,

$$h(k) = \begin{cases} b_k, & 0 \le k \le M \\ 0, & \text{otherwise} \end{cases} \tag{2.5.12}$$

If we return to (2.5.6) and set $M = 0$, the general linear time-invariant system reduces to a "purely recursive" system described by the difference equation

$$y(n) = -\sum_{k=1}^{N} a_k y(n-k) + b_0 x(n) \tag{2.5.13}$$

In this case the system output is a weighted linear combination of N past outputs and the present input.

Linear time-invariant systems described by a second-order difference equation are an important subclass of the more general systems described by (2.5.6) or (2.5.10) or (2.5.13). The reason for their importance will be explained later when we discuss quantization effects. Suffice to say at this point that second-order systems are usually used as basic building blocks for realizing higher-order systems.

The most general second-order system is described by the difference equation

$$\begin{aligned} y(n) = & -a_1 y(n-1) - a_2 y(n-2) + b_0 x(n) \\ & + b_1 x(n-1) + b_2 x(n-2) \end{aligned} \tag{2.5.14}$$

which is obtained from (2.5.6) by setting $N = 2$ and $M = 2$. The direct form II structure for realizing this system is shown in Fig. 2.35a. If we set $a_1 = a_2 = 0$, then (2.5.14) reduces to

$$y(n) = b_0 x(n) + b_1 x(n-1) + b_2 x(n-2) \tag{2.5.15}$$

which is a special case of the FIR system described by (2.5.11). The structure for realizing this system is shown in Fig. 2.35b. Finally, if we set $b_1 = b_2 = 0$ in (2.5.14), we obtain the purely recursive second-order system described by the difference equation

$$y(n) = -a_1 y(n-1) - a_2 y(n-2) + b_0 x(n) \tag{2.5.16}$$

which is a special case of (2.5.13). The structure for realizing this system is shown in Fig. 2.35c.

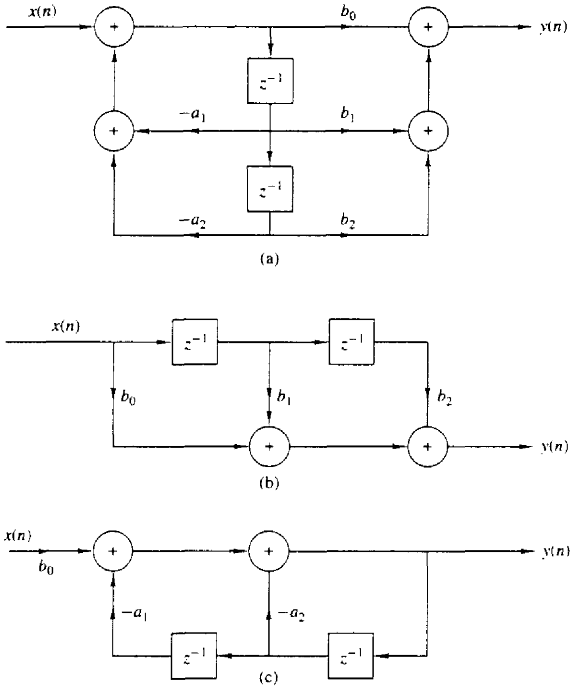


Figure 2.35 Structures for the realization of second-order systems: (a) general second-order system; (b) FIR system; (c) "purely recursive system"

2.5.2 Recursive and Nonrecursive Realizations of FIR Systems

We have already made the distinction between FIR and IIR systems, based on whether the impulse response $h(n)$ of the system has a finite duration, or an infinite duration. We have also made the distinction between recursive and nonrecursive systems. Basically, a causal recursive system is described by an input–output equation of the form

$$y(n) = F[y(n-1), \ldots, y(n-N), x(n), \ldots, x(n-M)] \tag{2.5.17}$$

and for a linear time-invariant system specifically, by the difference equation

$$y(n) = -\sum_{k=1}^{N} a_k y(n-k) + \sum_{k=0}^{M} b_k x(n-k) \tag{2.5.18}$$

On the other hand, causal nonrecursive systems do not depend on past values of the output and hence are described by an input–output equation of the form

$$y(n) = F[x(n), x(n-1), \ldots, x(n-M)] \tag{2.5.19}$$

and for linear time-invariant systems specifically, by the difference equation in (2.5.18) with $a_k = 0$ for $k = 1, 2, \ldots, N$.

In the case of FIR systems, we have already observed that it is always possible to realize such systems nonrecursively. In fact, with $a_k = 0$, $k = 1, 2, \ldots, N$, in (2.5.18), we have a system with an input–output equation

$$y(n) = \sum_{k=0}^{M} b_k x(n-k) \tag{2.5.20}$$

This is a nonrecursive and FIR system. As indicated in (2.5.12), the impulse response of the system is simply equal to the coefficients $\{b_k\}$. Hence every FIR system can be realized nonrecursively. On the other hand, any FIR system can also be realized recursively. Although the general proof of this statement is given later, we shall give a simple example to illustrate the point.

Suppose that we have an FIR system of the form

$$y(n) = \frac{1}{M+1} \sum_{k=0}^{M} x(n-k) \tag{2.5.21}$$

for computing the *moving average* of a signal $x(n)$. Clearly, this system is FIR with impulse response

$$h(n) = \frac{1}{M+1} \qquad 0 \le n \le M$$

Figure 2.36 illustrates the structure of the nonrecursive realization of the system. Now, suppose that we express (2.5.21) as

$$\begin{aligned} y(n) &= \frac{1}{M+1} \sum_{k=0}^{M} x(n-1-k) \\ &\quad + \frac{1}{M+1}[x(n) - x(n-1-M)] \\ &= y(n-1) + \frac{1}{M+1}[x(n) - x(n-1-M)] \end{aligned} \tag{2.5.22}$$

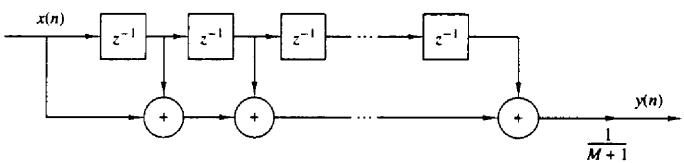


Figure 2.36 Nonrecursive realization of an FIR moving average system.

Now, (2.5.22) represents a recursive realization of the FIR system. The structure of this recursive realization of the moving average system is illustrated in Fig. 2.37.

In summary, we can think of the terms FIR and IIR as general characteristics that distinguish a type of linear time-invariant system, and of the terms *recursive* and *nonrecursive* as descriptions of the structures for realizing or implementing the system.

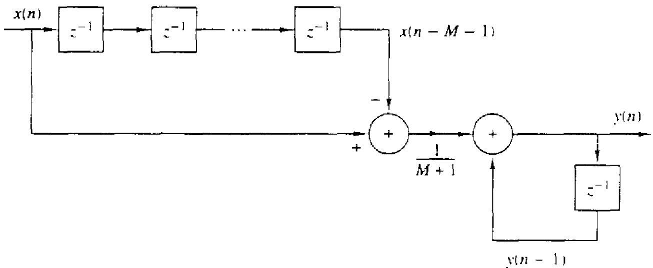


Figure 2.37 Recursive realization of an FIR moving average system.

2.6 CORRELATION OF DISCRETE-TIME SIGNALS

A mathematical operation that closely resembles convolution is correlation. Just as in the case of convolution, two signal sequences are involved in correlation. In contrast to convolution, however, our objective in computing the correlation between the two signals is to measure the degree to which the two signals are similar and thus to extract some information that depends to a large extent on the application. Correlation of signals is often encountered in radar, sonar, digital communications, geology, and other areas in science and engineering.

To be specific, let us suppose that we have two signal sequences $x(n)$ and $y(n)$ that we wish to compare. In radar and active sonar applications, $x(n)$ can represent the sampled version of the transmitted signal and $y(n)$ can represent the sampled version of the received signal at the output of the analog-to-digital (A/D) converter. If a target is present in the space being searched by the radar or sonar, the received signal $y(n)$ consists of a delayed version of the transmitted signal, reflected from the target, and corrupted by additive noise. Figure 2.38 depicts the radar signal reception problem.

We can represent the received signal sequence as

$$y(n) = \alpha x(n-D) + w(n) \tag{2.6.1}$$

where α is some attenuation factor representing the signal loss involved in the round-trip transmission of the signal $x(n)$, D is the round-trip delay, which is

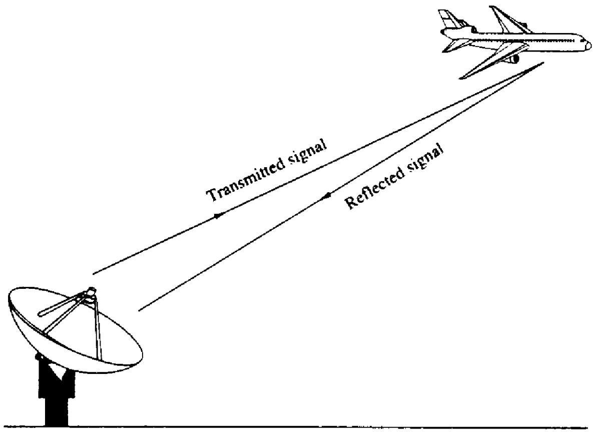


Figure 2.38 Radar target detection.

assumed to be an integer multiple of the sampling interval, and $w(n)$ represents the additive noise that is picked up by the antenna and any noise generated by the electronic components and amplifiers contained in the front end of the receiver. On the other hand, if there is no target in the space searched by the radar and sonar, the received signal $y(n)$ consists of noise alone.

Having the two signal sequences, $x(n)$, which is called the reference signal or transmitted signal, and $y(n)$, the received signal, the problem in radar and sonar detection is to compare $y(n)$ and $x(n)$ to determine if a target is present and, if so, to determine the time delay D and compute the distance to the target. In practice, the signal $x(n-D)$ is heavily corrupted by the additive noise to the point where a visual inspection of $y(n)$ does not reveal the presence or absence of the desired signal reflected from the target. Correlation provides us with a means for extracting this important information from $y(n)$.

Digital communications is another area where correlation is often used. In digital communications the information to be transmitted from one point to another is usually converted to binary from, that is, a sequence of zeros and ones, which are then transmitted to the intended receiver. To transmit a 0 we can transmit the signal sequence $x_0(n)$ for $0 \le n \le L-1$, and to transmit a 1 we can transmit the signal sequence $x_1(n)$ for $0 \le n \le L-1$, where L is some integer that denotes the number of samples in each of the two sequences. Very often, $x_1(n)$ is selected to be the negative of $x_0(n)$. The signal received by the intended receiver may be represented as

$$y(n) = x_i(n) + w(n) \qquad i = 0, 1 \qquad 0 \le n \le L-1 \tag{2.6.2}$$

where now the uncertainty is whether $x_0(n)$ or $x_1(n)$ is the signal component in $y(n)$, and $w(n)$ represents the additive noise and other interference inherent in any communication system. Again, such noise has its origin in the electronic components contained in the front end of the receiver. In any case, the receiver knows the possible transmitted sequences $x_0(n)$ and $x_1(n)$ and is faced with the task of comparing the received signal $y(n)$ with both $x_0(n)$ and $x_1(n)$ to determine which of the two signals better matches $y(n)$. This comparison process is performed by means of the correlation operation described in the following subsection.

2.6.1 Crosscorrelation and Autocorrelation Sequences

Suppose that we have two real signal sequences $x(n)$ and $y(n)$ each of which has finite energy. The *crosscorrelation* of $x(n)$ and $y(n)$ is a sequence $r_{xy}(l)$, which is defined as

$$r_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n)y(n-l) \qquad l = 0, \pm1, \pm2, \ldots \tag{2.6.3}$$

or, equivalently, as

$$r_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n+l)y(n) \qquad l = 0, \pm1, \pm2, \ldots \tag{2.6.4}$$

The index l is the (time) shift (or *lag*) parameter and the subscripts xy on the crosscorrelation sequence $r_{xy}(l)$ indicate the sequences being correlated. The order of the subscripts, with x preceding y, indicates the direction in which one sequence is shifted, relative to the other. To elaborate, in (2.6.3), the sequence $x(n)$ is left unshifted and $y(n)$ is shifted by l units in time, to the right for l positive and to the left for l negative. Equivalently, in (2.6.4), the sequence $y(n)$ is left unshifted and $x(n)$ is shifted by l units in time, to the left for l positive and to the right for l negative. But shifting $x(n)$ to the left by l units relative to $y(n)$ is equivalent to shifting $y(n)$ to the right by l units relative to $x(n)$. Hence the computations (2.6.3) and (2.6.4) yield identical crosscorrelation sequences.

If we reverse the roles of $x(n)$ and $y(n)$ in (2.6.3) and (2.6.4) and therefore reverse the order of the indices xy, we obtain the crosscorrelation sequence

$$r_{yx}(l) = \sum_{n=-\infty}^{\infty} y(n)x(n-l) \tag{2.6.5}$$

or, equivalently,

$$r_{yx}(l) = \sum_{n=-\infty}^{\infty} y(n+l)x(n) \tag{2.6.6}$$

By comparing (2.6.3) with (2.6.6) or (2.6.4) with (2.6.5), we conclude that

$$r_{xy}(l) = r_{yx}(-l) \tag{2.6.7}$$

Therefore, $r_{yx}(l)$ is simply the folded version of $r_{xy}(l)$, where the folding is done with respect to $l = 0$. Hence, $r_{yx}(l)$ provides exactly the same information as $r_{xy}(l)$, with respect to the similarity of $x(n)$ to $y(n)$.

Example 2.6.1

Determine the crosscorrelation sequence $r_{xy}(l)$ of the sequences

$$x(n) = \{\ldots, 0, 0, 2, -1, 3, 7, \underset{\uparrow}{1}, 2, -3, 0, 0, \ldots\}$$

$$y(n) = \{\ldots, 0, 0, 1, -1, 2, -2, \underset{\uparrow}{4}, 1, -2, 5, 0, 0, \ldots\}$$

Solution Let us use the definition in (2.6.3) to compute $r_{xy}(l)$. For $l = 0$ we have

$$r_{xy}(0) = \sum_{n=-\infty}^{\infty} x(n)y(n)$$

The product sequence $v_0(n) = x(n)y(n)$ is

$$v_0(n) = \{\ldots, 0, 0, 2, 1, 6, -14, \underset{\uparrow}{4}, 2, 6, 0, 0, \ldots\}$$

and hence the sum over all values of n is

$$r_{xy}(0) = 7$$

For $l > 0$, we simply shift $y(n)$ to the right relative to $x(n)$ by l units, compute the product sequence $v_l(n) = x(n)y(n-l)$, and finally, sum over all values of the product sequence. Thus we obtain

$$r_{xy}(1) = 13, \qquad r_{xy}(2) = -18, \qquad r_{xy}(3) = 16, \qquad r_{xy}(4) = -7$$

$$r_{xy}(5) = 5, \qquad r_{xy}(6) = -3, \qquad r_{xy}(l) = 0, \qquad l \geq 7$$

For $l < 0$, we shift $y(n)$ to the left relative to $x(n)$ by l units, compute the product sequence $v_l(n) = x(n)y(n-l)$, and sum over all values of the product sequence. Thus we obtain the values of the crosscorrelation sequence

$$r_{xy}(-1) = 0, \qquad r_{xy}(-2) = 33, \qquad r_{xy}(-3) = -14, \qquad r_{xy}(-4) = 36$$

$$r_{xy}(-5) = 19, \qquad r_{xy}(-6) = -9, \qquad r_{xy}(-7) = 10, \qquad r_{xy}(l) = 0, \ l \leq -8$$

Therefore, the crosscorrelation sequence of $x(n)$ and $y(n)$ is

$$r_{xy}(l) = \{10, -9, 19, 36, -14, 33, 0, \underset{\uparrow}{7}, 13, -18, 16, -7, 5, -3\}$$

The similarities between the computation of the crosscorrelation of two sequences and the convolution of two sequences is apparent. In the computation of convolution, one of the sequences is folded, then shifted, then multiplied by the other sequence to form the product sequence for that shift, and finally, the values of the product sequence are summed. Except for the folding operation, the computation of the crosscorrelation sequence involves the same operations: shifting one of the sequences, multiplication of the two sequences, and summing over all values of the product sequence. Consequently, if we have a computer program that performs convolution, we can use it to perform crosscorrelation by providing as inputs to the program, the sequence $x(n)$ and the folded sequence $y(-n)$. Then the convolution of $x(n)$ with $y(-n)$ yields the crosscorrelation $r_{xy}(l)$, that is,

$$r_{xy}(l) = x(l) * y(-l) \tag{2.6.8}$$

In the special case where $y(n) = x(n)$, we have the *autocorrelation* of $x(n)$, which is defined as the sequence

$$r_{xx}(l) = \sum_{n=-\infty}^{\infty} x(n)x(n-l) \tag{2.6.9}$$

or, equivalently, as

$$r_{xx}(l) = \sum_{n=-\infty}^{\infty} x(n+l)x(n) \tag{2.6.10}$$

In dealing with finite-duration sequences, it is customary to express the autocorrelation and crosscorrelation in terms of the finite limits on the summation. In particular, if $x(n)$ and $y(n)$ are causal sequences of length N [i.e., $x(n) = y(n) = 0$ for $n < 0$ and $n \geq N$], the crosscorrelation and autocorrelation sequences may be expressed as

$$r_{xy}(l) = \sum_{n=i}^{N-|k|-1} x(n)y(n-l) \tag{2.6.11}$$

and

$$r_{xx}(l) = \sum_{n=i}^{N-|k|-1} x(n)x(n-l) \tag{2.6.12}$$

where $i = l$, $k = 0$ for $l \geq 0$, and $i = 0$, $k = l$ for $l < 0$.

2.6.2 Properties of the Autocorrelation and Crosscorrelation Sequences

The autocorrelation and crosscorrelation sequences have a number of important properties that we now present. To develop these properties, let us assume that we have two sequences $x(n)$ and $y(n)$ with finite energy from which we form the linear combination,

$$ax(n) + by(n-l)$$

where a and b are arbitrary constants and l is some time shift. The energy in this signal is

$$\begin{aligned}\sum_{n=-\infty}^{\infty} [ax(n) + by(n-l)]^2 &= a^2 \sum_{n=-\infty}^{\infty} x^2(n) + b^2 \sum_{n=-\infty}^{\infty} y^2(n-l) \\ &\quad + 2ab \sum_{n=-\infty}^{\infty} x(n)y(n-l) \\ &= a^2 r_{xx}(0) + b^2 r_{yy}(0) + 2ab r_{xy}(l)\end{aligned} \tag{2.6.13}$$

First, we note that $r_{xx}(0) = E_x$ and $r_{yy}(0) = E_y$, which are the energies of $x(n)$ and $y(n)$, respectively. It is obvious that

$$a^2 r_{xx}(0) + b^2 r_{yy}(0) + 2ab r_{xy}(l) \geq 0 \tag{2.6.14}$$

Now, assuming that $b \neq 0$, we can divide (2.6.14) by b^2 to obtain

$$r_{xx}(0)\left(\frac{a}{b}\right)^2 + 2r_{xy}(l)\left(\frac{a}{b}\right) + r_{yy}(0) \geq 0$$

We view this equation as a quadratic with coefficients $r_{xx}(0)$, $2r_{xy}(l)$, and $r_{yy}(0)$. Since the quadratic is nonnegative, it follows that the discriminant of this quadratic must be nonpositive, that is,

$$4[r_{xy}^2(l) - r_{xx}(0)r_{yy}(0)] \leq 0$$

Therefore, the crosscorrelation sequence satisfies the condition that

$$|r_{xy}(l)| \leq \sqrt{r_{xx}(0)r_{yy}(0)} = \sqrt{E_x E_y} \tag{2.6.15}$$

In the special case where $y(n) = x(n)$, (2.6.15) reduces to

$$|r_{xx}(l)| \leq r_{xx}(0) = E_x \tag{2.6.16}$$

This means that the autocorrelation sequence of a signal attains its maximum value at zero lag. This result is consistent with the notion that a signal matches perfectly with itself at zero shift. In the case of the crosscorrelation sequence, the upper bound on its values is given in (2.6.15).

Note that if any one or both of the signals involved in the crosscorrelation are scaled, the shape of the crosscorrelation sequence does not change, only the amplitudes of the crosscorrelation sequence are scaled accordingly. Since scaling is unimportant, it is often desirable, in practice, to normalize the autocorrelation and crosscorrelation sequences to the range from -1 to 1. In the case of the autocorrelation sequence, we can simply divide by $r_{xx}(0)$. Thus the normalized autocorrelation sequence is defined as

$$\rho_{xx}(l) = \frac{r_{xx}(l)}{r_{xx}(0)} \tag{2.6.17}$$

Similarly, we define the normalized crosscorrelation sequence

$$\rho_{xy}(l) = \frac{r_{xy}(l)}{\sqrt{r_{xx}(0)r_{yy}(0)}} \tag{2.6.18}$$

Now $|\rho_{xx}(l)| \leq 1$ and $|\rho_{xy}(l)| \leq 1$, and hence these sequences are independent of signal scaling.

Finally, as we have already demonstrated, the crosscorrelation sequence satisfies the property

$$r_{xy}(l) = r_{yx}(-l)$$

With $y(n) = x(n)$, this relation results in the following important property for the autocorrelation sequence

$$r_{xx}(l) = r_{xx}(-l) \tag{2.6.19}$$

Hence the autocorrelation function is an even function. Consequently, it suffices to compute $r_{xx}(l)$ for $l \geq 0$.

Example 2.6.2

Compute the autocorrelation of the signal

$$x(n) = a^n u(n),\ 0 < a < 1$$

Solution Since $x(n)$ is an infinite-duration signal, its autocorrelation also has infinite duration. We distinguish two cases.

If $l \geq 0$, from Fig. 2.39 we observe that

$$r_{xx}(l) = \sum_{n=l}^{\infty} x(n)x(n-l) = \sum_{n=l}^{\infty} a^n a^{n-l} = a^{-l} \sum_{n=l}^{\infty} (a^2)^n$$

Since $a < 1$, the infinite series con erges and we obtain

$$r_{xx}(l) = \frac{1}{1-a^2} a^l \qquad l \geq 0$$

For $l < 0$ we have

$$r_{xx}(l) = \sum_{n=0}^{\infty} x(n)x(n-l) = a^{-l} \sum_{n=0}^{\infty} (a^2)^n = \frac{1}{1-a^2} a^{-l} \qquad l < 0$$

But when l is negative, $a^{-l} = a^{|l|}$. Thus the two relations for $r_{xx}(l)$ can be combined into the following expression:

$$r_{xx}(l) = \frac{1}{1-a^2} a^{|l|} \qquad -\infty < l < \infty \tag{2.6.20}$$

The sequence $r_{xx}(l)$ is shown in Fig. 2.42(d). We observe that

$$r_{xx}(-l) = r_{xx}(l)$$

and

$$r_{xx}(0) = \frac{1}{1-a^2}$$

Therefore, the normalized autocorrelation sequence is

$$\rho_{xx}(l) = \frac{r_{xx}(l)}{r_{xx}(0)} = a^{|l|} \qquad -\infty < l < \infty \tag{2.6.21}$$

2.6.3 Correlation of Periodic Sequences

In Section 2.6.1 we defined the crosscorrelation and autocorrelation sequences of energy signals. In this section we consider the correlation sequences of power signals and, in particular, periodic signals.

Let $x(n)$ and $y(n)$ be two power signals. Their crosscorrelation sequence is defined as

$$r_{xy}(l) = \lim_{M\to\infty} \frac{1}{2M+1} \sum_{n=-M}^{M} x(n)y(n-l) \tag{2.6.22}$$

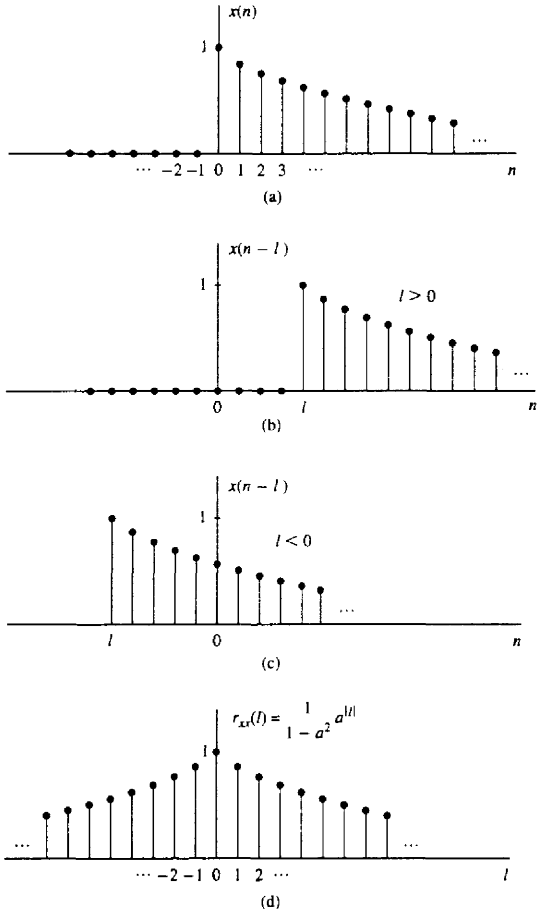


Figure 2.39 Computation of the autocorrelation of the signal $x(n) = a^n$, $0 < a < 1$.

If $x(n) = y(n)$, we have the definition of the autocorrelation sequence of a power signal as

$$r_{xx}(l) = \lim_{M\to\infty} \frac{1}{2M+1} \sum_{n=-M}^{M} x(n)x(n-l) \tag{2.6.23}$$

In particular, if $x(n)$ and $y(n)$ are two periodic sequences, each with period N, the averages indicated in (2.6.22) and (2.6.23) over the infinite interval, are identical to the averages over a single period, so that (2.6.22) and (2.6.23) reduce to

$$r_{xy}(l) = \frac{1}{N}\sum_{n=0}^{N-1} x(n)y(n-l) \tag{2.6.24}$$

and

$$r_{xx}(l) = \frac{1}{N}\sum_{n=0}^{N-1} x(n)x(n-l) \tag{2.6.25}$$

It is clear that $r_{xy}(l)$ and $r_{xx}(l)$ are periodic correlation sequences with period N. The factor $1/N$ can be viewed as a normalization scale factor.

In some practical applications, correlation is used to identify periodicities in an observed physical signal which may be corrupted by random interference. For example, consider a signal sequence $y(n)$ of the form

$$y(n) = x(n) + w(n) \tag{2.6.26}$$

where $x(n)$ is a periodic sequence of some unknown period N and $w(n)$ represents an additive random interference. Suppose that we observe M samples of $y(n)$, say $0 \le n \le M-1$, where $M >> N$. For all practical purposes, we can assume that $y(n) = 0$ for $n < 0$ and $n \ge M$. Now the autocorrelation sequence of $y(n)$, using the normalization factor of $1/M$, is

$$r_{yy}(l) = \frac{1}{M}\sum_{n=0}^{M-1} y(n)y(n-l) \tag{2.6.27}$$

If we substitute for $y(n)$ from (2.6.26) into (2.6.27) we obtain

$$\begin{aligned} r_{yy}(l) &= \frac{1}{M}\sum_{n=0}^{M-1}[x(n)+w(n)][x(n-l)+w(n-l)] \\ &= \frac{1}{M}\sum_{n=0}^{M-1} x(n)x(n-l) \\ &\quad + \frac{1}{M}\sum_{n=0}^{M-1}[x(n)w(n-l) + w(n)x(n-l)] \\ &\quad + \frac{1}{M}\sum_{n=0}^{M-1} w(n)w(n-l) \\ &= r_{xx}(l) + r_{xw}(l) + r_{wx}(l) + r_{ww}(l) \end{aligned} \tag{2.6.28}$$

The first factor on the right-hand side of (2.6.28) is the autocorrelation sequence of $x(n)$. Since $x(n)$ is periodic, its autocorrelation sequence exhibits the same periodicity, thus containing relatively large peaks at $l = 0, N, 2N$, and so on. However, as the shift l approaches M, the peaks are reduced in amplitude due to the fact that we have a finite data record of M samples so that many of the products $x(n)x(n-l)$ are zero. Consequently, we should avoid computing $r_{yy}(l)$ for large lags, say, $l > M/2$.

The crosscorrelations $r_{xw}(l)$ and $r_{wx}(l)$ between the signal $x(n)$ and the additive random interference are expected to be relatively small as a result of the expectation that $x(n)$ and $w(n)$ will be totally unrelated. Finally, the last term on the right-hand side of (2.6.28) is the autocorrelation sequence of the random sequence $w(n)$. This correlation sequence will certainly contain a peak at $l = 0$, but because of its random characteristics, $r_{ww}(l)$ is expected to decay rapidly toward zero. Consequently, only $r_{xx}(l)$ is expected to have large peaks for $l > 0$. This behavior allows us to detect the presence of the periodic signal $x(n)$ buried in the interference $w(n)$ and to identify its period.

An example that illustrates the use of autocorrelation to identify a hidden periodicity in an observed physical signal is shown in Fig. 2.40. This figure illustrates the autocorrelation (normalized) sequence for the Wölfer sunspot numbers for $0 \leq l \leq 20$, where any value of l corresponds to one year. These numbers are given in Table 2.2 for the 100-year period 1770–1869. There is clear evidence in this figure that a periodic trend exists, with a period of 10 to 11 years.

Example 2.6.3

Suppose that a signal sequence $x(n) = \sin(\pi/5)n$, for $0 \leq n \leq 99$ is corrupted by an additive noise sequence $w(n)$, where the values of the additive noise are selected independently from sample to sample, from a uniform distribution over the range

TABLE 2.2 YEARLY WÖLFER SUNSPOT NUMBERS

1770	101	1795	21	1820	16	1845	40
1771	82	1796	16	1821	7	1846	62
1772	66	1797	6	1822	4	1847	98
1773	35	1798	4	1823	2	1848	124
1774	31	1799	7	1824	8	1849	96
1775	7	1800	14	1825	17	1850	66
1776	20	1801	34	1826	36	1851	64
1777	92	1802	45	1827	50	1852	54
1778	154	1803	43	1828	62	1853	39
1779	125	1804	48	1829	67	1854	21
1780	85	1805	42	1830	71	1855	7
1781	68	1806	28	1831	48	1856	4
1782	38	1807	10	1832	28	1857	23
1783	23	1808	8	1833	8	1858	55
1784	10	1809	2	1834	13	1859	94
1785	24	1810	0	1835	57	1860	96
1786	83	1811	1	1836	122	1861	77
1787	132	1812	5	1837	138	1862	59
1788	131	1813	12	1838	103	1863	44
1789	118	1814	14	1839	86	1864	47
1790	90	1815	35	1840	63	1865	30
1791	67	1816	46	1841	37	1866	16
1792	60	1817	41	1842	24	1867	7
1793	47	1818	30	1843	11	1868	37
1794	41	1819	24	1844	15	1869	74

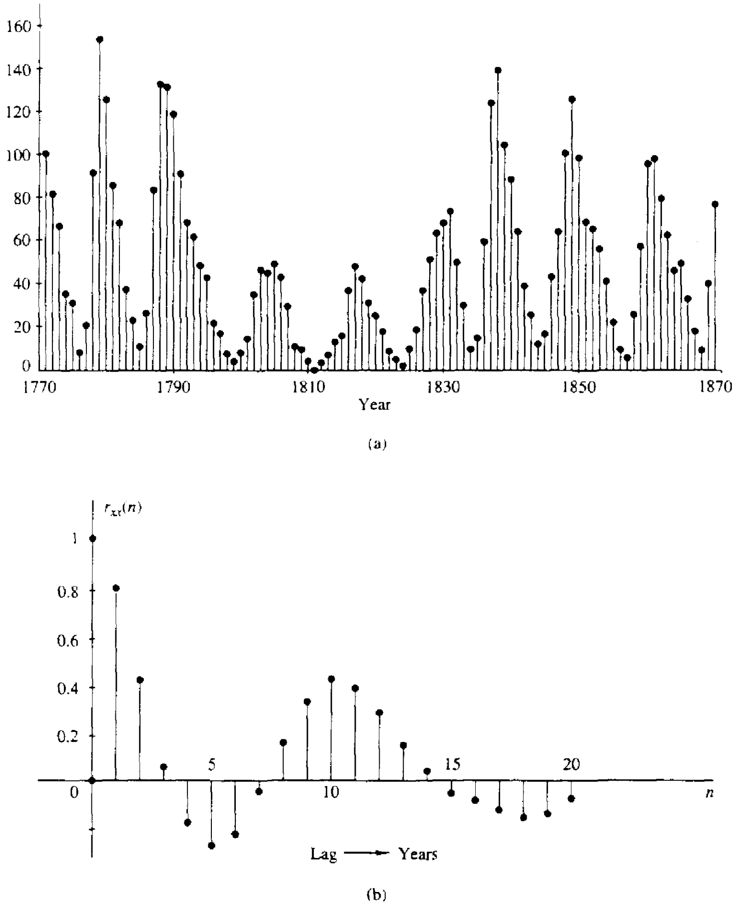


Figure 2.40 Identification of periodicity in the Wölfer sunspot numbers: (a) annual Wölfer sunspot numbers; (b) autocorrelation sequence.

$(-\Delta/2, \Delta/2)$, where Δ is a parameter of the distribution. The observed sequence is $y(n) = x(n)+w(n)$. Determine the autocorrelation sequence $r_{yy}(n)$ and thus determine the period of the signal $x(n)$.

Solution The assumption is that the signal sequence $x(n)$ has some unknown period that we are attempting to determine from the noise-corrupted observations $\{y(n)\}$. Although $x(n)$ is periodic with period 10, we have only a finite-duration sequence of length $M = 100$ [i.e., 10 periods of $x(n)$]. The noise power level P_w in the sequence $w(n)$ is determined by the parameter Δ. We simply state that $P_w = \Delta^2/12$. The signal power level is $P_x = \frac{1}{2}$. Therefore, the signal-to-noise ratio (SNR) is defined as

$$\frac{P_x}{P_w} = \frac{\frac{1}{2}}{\Delta^2/12} = \frac{6}{\Delta^2}$$

Usually, the SNR is expressed on a logarithmic scale in decibels (dB) as $10\log_{10}(P_x/P_w)$.

Figure 2.41 illustrates a sample of a noise sequence $w(n)$, and the observed sequence $y(n) = x(n) + w(n)$ when the SNR = 1 dB. The autocorrelation sequence

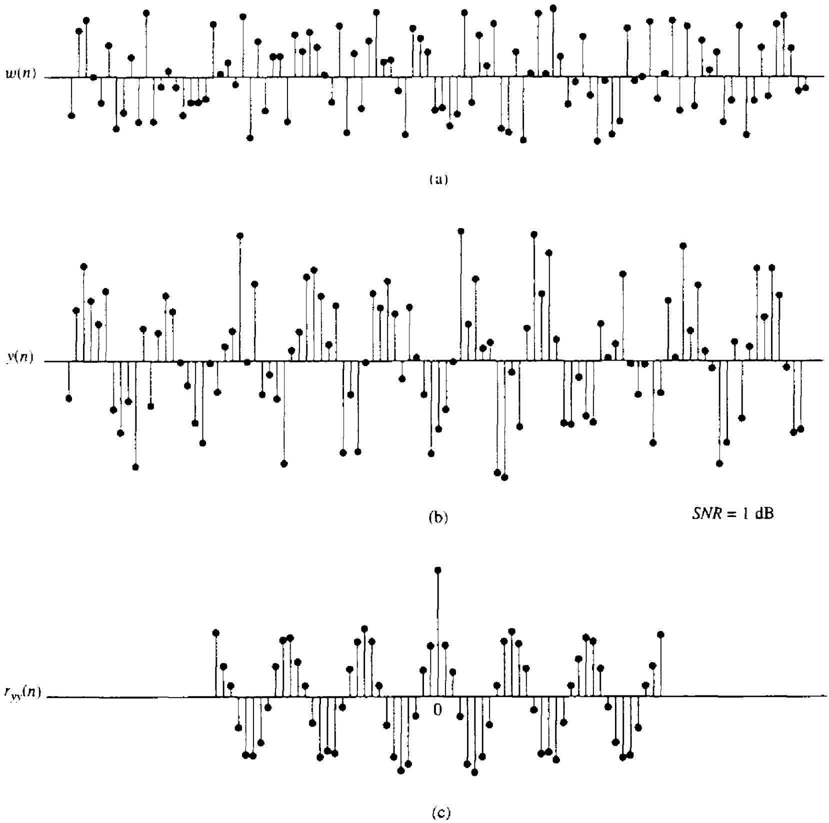


Figure 2.41 Use of autocorrelation to detect the presence of a periodic signal corrupted by noise.

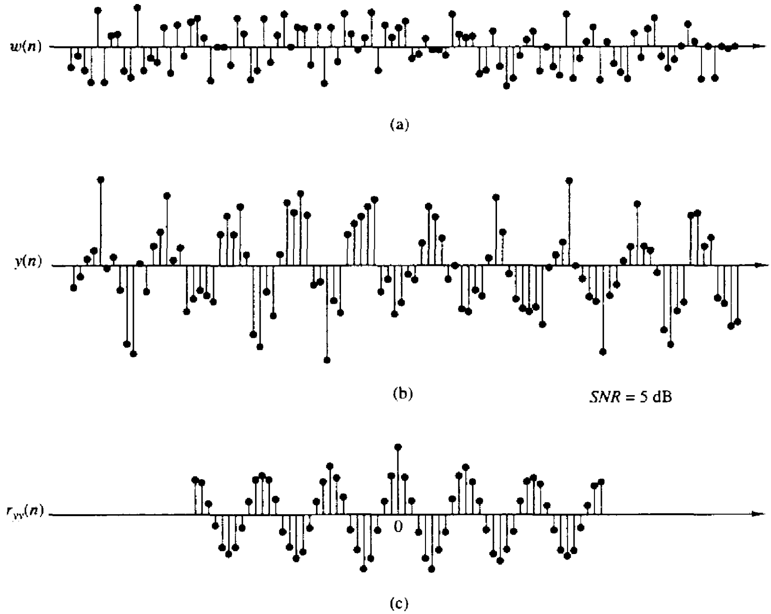


Figure 2.42 Use of autocorrelation to detect the presence of a periodic signal corrupted by noise.

$r_{yy}(l)$ is illustrated in Fig. 2.41c. We observe that the periodic signal $x(n)$, embedded in $y(n)$, results in a periodic autocorrelation function $r_{xx}(l)$ with period $N = 10$. The effect of the additive noise is to add to the peak value at $l = 0$, but for $l \neq 0$, the correlation sequence $r_{ww}(l) \approx 0$ as a result of the fact that values of $w(n)$ were generated independently. Such noise is usually called *white noise.* The presence of this noise explains the reason for the large peak at $l = 0$. The smaller, nearly equal peaks at $l = \pm 10, \pm 20, \ldots$ are due the periodic characteristics of $x(n)$.

Figure 2.42 illustrates the noise sequence $w(n)$, the noise-corrupted signal $y(n)$, and the autocorrelation sequence $r_{yy}(l)$ for the same signal, within which is embedded a signal at a smaller noise level. In this case, the SNR = 5 dB. Even with this relatively small noise level, the periodicity of the signal is not easily determined from observation of $y(n)$. However, it is clearly evident from observation of the autocorrelation sequence $r_{yy}(n)$.

2.6.4 Computation of Correlation Sequences

As indicated on Section 2.6.1, the procedure for computing the crosscorrelation sequence between $x(n)$ and $y(n)$ involves shifting one of the sequences, say $x(n)$, to obtain $x(n-l)$, multiplying the shifted sequence by $y(n)$ to obtain the product sequence $y(n)x(n-l)$, and then summing all the values of the product sequence to obtain $r_{yx}(l)$. This procedure is repeated for different values of the lag l. Except for the folding operation that is involved in convolution, these basic operations for computing the correlation sequence are identical to those in convolution.

The procedure for computing the convolution is directly applicable to computing the correlation of two sequences. Specifically, if we fold $y(n)$ to obtain $y(-n)$, then the convolution of $x(n)$ with $y(-n)$ is identical to the crosscorrelation of $x(n)$ with $y(n)$. That is,

$$r_{xy}(l) = x(n) * y(-n)|_{n=l} \tag{2.6.29}$$

As a consequence, the computational procedure described for convolution can be applied directly to the computation of the correlation sequence.

We now describe an algorithm that can be easily programmed to compute the crosscorrelation sequence of two finite-duration signals $x(n)$, $0 \le n \le N-1$, and $y(n)$, $0 \le n \le M-1$.

The algorithm computes $r_{xy}(l)$ for positive lags. According to the relation $r_{xy}(-l) = r_{yx}(l)$, the values of $r_{xy}(l)$ for negative lags can be obtained by using the same algorithm for positive lags, and interchanging the roles of $x(n)$ and $y(n)$. We observe that if $M \le N$, $r_{xy}(l)$ can be computed by the relations

$$r_{xy}(l) = \begin{cases} \displaystyle\sum_{n=l}^{M-1+l} x(n)y(n-l), & 0 \le l \le N-M \\ \displaystyle\sum_{n=l}^{N-1} x(n)y(n-l), & N-M < l \le N-1 \end{cases} \tag{2.6.30}$$

On the other hand, if $M > N$, the formula for the crosscorrelation becomes

$$r_{xy}(l) = \sum_{n=l}^{N-1} x(n)y(n-l) \qquad 0 \le l \le N-1 \tag{2.6.31}$$

The formulas in (2.6.30) and (2.6.31) can be combined and computed by means of the following simple algorithm illustrated in the flowchart in Fig. 2.43. By interchanging the roles of $x(n)$ and $y(n)$ and recomputing the crosscorrelation sequence, we obtain the values of $r_{xy}(l)$ corresponding to negative shifts l.

If we wish to compute the autocorrelation sequence $r_{xx}(l)$, we set $y(n) = x(n)$ and $M = N$ in (2.6.31). The computation of $r_{xx}(l)$ can be done by means of the same algorithm for positive shifts only.

2.6.5 Input–Output Correlation Sequences

In this section we derive two input–output relationships for LTI systems in the "correlation domain." Let us assume that a signal $x(n)$ with known autocorrelation $r_{xx}(l)$ is applied to an LTI system with impulse response $h(n)$, producing the

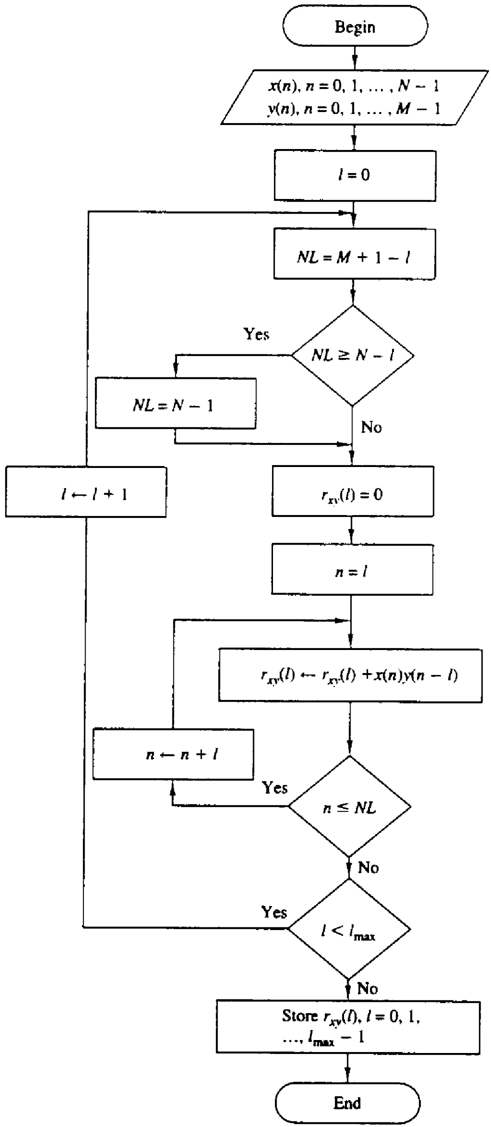


Figure 2.43 Flowchart for software implementation of crosscorrelation.

output signal

$$y(n) = h(n) * x(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k)$$

The crosscorrelation between the output and the input signal is

$$r_{yx}(l) = y(l) * x(-l) = h(l) * [x(l) * x(-l)]$$

or

$$r_{yx}(l) = h(l) * r_{xx}(l) \tag{2.6.32}$$

where we have used (2.6.8) and the properties of convolution. Hence the crosscorrelation between the input and the output of the system is the convolution of the impulse response with the autocorrelation of the input sequence. Alternatively, $r_{yx}(l)$ may be viewed as the output of the LTI system when the input sequence is $r_{xx}(l)$. This is illustrated in Fig. 2.44. If we replace l by $-l$ in (2.6.32), we obtain

$$r_{xy}(l) = h(-l) * r_{xx}(l)$$

The autocorrelation of the output signal can be obtained by using (2.6.8) with $x(n) = y(n)$ and the properties of convolution. Thus we have

$$\begin{aligned} r_{yy}(l) &= y(l) * y(-l) \\ &= [h(l) * x(l)] * [h(-l) * x(-l)] \\ &= [h(l) * h(-l)] * [x(l) * x(-l)] \\ &= r_{hh}(l) * r_{xx}(l) \end{aligned} \tag{2.6.33}$$

The autocorrelation $r_{hh}(l)$ of the impulse response $h(n)$ exists if the system is stable. Furthermore, the stability insures that the system does not change the type (energy or power) of the input signal. By evaluating (2.6.33) for $l = 0$ we obtain

$$r_{yy}(0) = \sum_{k=-\infty}^{\infty} r_{hh}(k) r_{xx}(k) \tag{2.6.34}$$

which provides the energy (or power) of the output signal in terms of autocorrelations. These relationships hold for both energy and power signals. The direct derivation of these relationships for energy and power signals, and their extensions to complex signals, are left as exercises for the student.

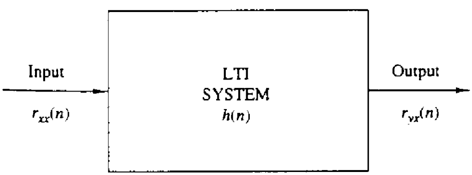


Figure 2.44 Input–output relation for crosscorrelation $r_{yx}(n)$.

2.7 SUMMARY AND REFERENCES

The major theme of this chapter is the characterization of discrete-time signals and systems in the time domain. Of particular importance is the class of linear time-invariant (LTI) systems which are widely used in the design and implementation of digital signal processing systems. We characterized LTI systems by their unit sample response $h(n)$ and derived the convolution summation, which is a formula for determining the response $y(n)$ of the system characterized by $h(n)$ to any given input sequence $x(n)$.

The class of LTI systems characterized by linear difference equations with constant coefficients is by far the most important of the LTI systems in the theory and application of digital signal processing. The general solution of a linear difference equation with constant coefficients was derived in this chapter and shown to consist of two components: the solution of the homogeneous equation which represents the natural response of the system when the input is zero, and the particular solution, which represents the response of the system to the input signal. From the difference equation, we also demonstrated how to derive the unit sample response of the LTI system.

Linear time-invariant systems were generally subdivided into FIR (finite-duration impulse response) and IIR (infinite-duration impulse response) depending on whether $h(n)$ has finite duration or infinite duration, respectively. The realizations of such systems were briefly described. Furthermore, in the realization of FIR systems, we made the distinction between recursive and nonrecursive realizations. On the other hand, we observed that IIR systems can be implemented recursively, only.

There are a number of texts on discrete-time signals and systems. We mention as examples the books by McGillem and Cooper (1984), Oppenheim and Willsky (1983), and Siebert (1986). Linear constant-coefficient difference equations are treated in depth in the books by Hildebrand (1952) and Levy and Lessman (1961).

The last topic in this chapter, on correlation of discrete-time signals, plays an important role in digital signal processing, especially in applications dealing with digital communications, radar detection and estimation, sonar, and geophysics. In our treatment of correlation sequences, we avoided the use of statistical concepts. Correlation is simply defined as a mathematical operation between two sequences, which produces another sequence, called either the *crosscorrelation sequence* when the two sequences are different, or the *autocorrelation sequence* when the two sequences are identical.

In practical applications in which correlation is used, one (or both) of the sequences is (are) contaminated by noise and, perhaps, by other forms of interference. In such a case, the noisy sequence is called a *random sequence* and is characterized in statistical terms. The corresponding correlation sequence becomes a function of the statistical characteristics of the noise and any other interference.

The statistical characterization of sequences and their correlation is treated in Appendix A. Supplementary reading on probabilistic and statistical concepts dealing with correlation can be found in the books by Davenport (1970), Helstrom (1990), Papoulis (1984), and Peebles (1987).

PROBLEMS

2.1 A discrete-time signal $x(n)$ is defined as

$$x(n) = \begin{cases} 1 + \frac{n}{3}, & -3 \le n \le -1 \\ 1, & 0 \le n \le 3 \\ 0, & \text{elsewhere} \end{cases}$$

(a) Determine its values and sketch the signal $x(n)$.

(b) Sketch the signals that result if we:

(1) First fold $x(n)$ and then delay the resulting signal by four samples.

(2) First delay $x(n)$ by four samples and then fold the resulting signal.

(c) Sketch the signal $x(-n+4)$.

(d) Compare the results in parts (b) and (c) and derive a rule for obtaining the signal $x(-n+k)$ from $x(n)$.

(e) Can you express the signal $x(n)$ in terms of signals $\delta(n)$ and $u(n)$?

2.2 A discrete-time signal $x(n)$ is shown in Fig. P2.2. Sketch and label carefully each of the following signals.

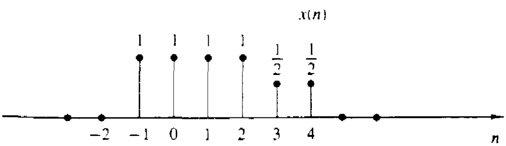


Figure P2.2

(a) $x(n-2)$ **(b)** $x(4-n)$ **(c)** $x(n+2)$ **(d)** $x(n)u(2-n)$

(e) $x(n-1)\delta(n-3)$ **(f)** $x(n^2)$ **(g)** even part of $x(n)$

(h) odd part of $x(n)$

2.3 Show that

(a) $\delta(n) = u(n) - u(n-1)$

(b) $u(n) = \sum_{k=-\infty}^{n} \delta(k) = \sum_{k=0}^{\infty} \delta(n-k)$

2.4 Show that any signal can be decomposed into an even and an odd component. Is the decomposition unique? Illustrate your arguments using the signal

$$x(n) = \{2, 3, \underset{\uparrow}{4}, 5, 6\}$$

2.5 Show that the energy (power) of a real-valued energy (power) signal is equal to the sum of the energies (powers) of its even and odd components.

2.6 Consider the system

$$y(n) = \mathcal{T}[x(n)] = x(n^2)$$

(a) Determine if the system is time invariant.

(b) To clarify the result in part (a) assume that the signal

$$x(n) = \begin{cases} 1, & 0 \le n \le 3 \\ 0, & \text{elsewhere} \end{cases}$$

is applied into the system.

(1) Sketch the signal $x(n)$.

(2) Determine and sketch the signal $y(n) = \mathcal{T}[x(n)]$.

(3) Sketch the signal $y_2'(n) = y(n-2)$.

(4) Determine and sketch the signal $x_2(n) = x(n-2)$.

(5) Determine and sketch the signal $y_2(n) = \mathcal{T}[x_2(n)]$.

(6) Compare the signals $y_2(n)$ and $y(n-2)$. What is your conclusion?

(c) Repeat part (b) for the system

$$y(n) = x(n) - x(n-1)$$

Can you use this result to make any statement about the time invariance of this system? Why?

(d) Repeat parts (b) and (c) for the system

$$y(n) = \mathcal{T}[x(n)] = nx(n)$$

2.7 A discrete-time system can be

(1) Static or dynamic

(2) Linear or nonlinear

(3) Time invariant or time varying

(4) Causal or noncausal

(5) Stable or unstable

Examine the following systems with respect to the properties above.

(a) $y(n) = \cos[x(n)]$

(b) $y(n) = \sum_{k=-\infty}^{n+1} x(k)$

(c) $y(n) = x(n)\cos(\omega_0 n)$

(d) $y(n) = x(-n+2)$

(e) $y(n) = \text{Trun}[x(n)]$, where $\text{Trun}[x(n)]$ denotes the integer part of $x(n)$, obtained by truncation

(f) $y(n) = \text{Round}[x(n)]$, where $\text{Round}[x(n)]$ denotes the integer part of $x(n)$ obtained by rounding

Remark: The systems in parts (e) and (f) are quantizers that perform truncation and rounding, respectively.

(g) $y(n) = |x(n)|$

(h) $y(n) = x(n)u(n)$

(i) $y(n) = x(n) + nx(n+1)$

(j) $y(n) = x(2n)$

(k) $y(n) = \begin{cases} x(n), & \text{if } x(n) \ge 0 \\ 0, & \text{if } x(n) < 0 \end{cases}$

(l) $y(n) = x(-n)$

(m) $y(n) = \text{sign}[x(n)]$

(n) The ideal sampling system with input $x_a(t)$ and output $x(n) = x_a(nT)$, $-\infty < n < \infty$

2.8 Two discrete-time systems $\mathcal{T}_1$ and $\mathcal{T}_2$ are connected in cascade to form a new system $\mathcal{T}$ as shown in Fig. P2.8. Prove or disprove the following statements.

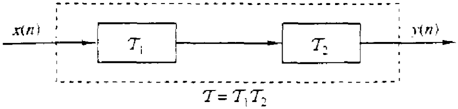


Figure P2.8

(a) If T_1 and T_2 are linear, then T is linear (i.e., the cascade connection of two linear systems is linear).
(b) If T_1 and T_2 are time invariant, then T is time invariant.
(c) If T_1 and T_2 are causal, then T is causal.
(d) If T_1 and T_2 are linear and time invariant, the same holds for T.
(e) If T_1 and T_2 are linear and time invariant, then interchanging their order does not change the system T.
(f) As in part (e) except that T_1, T_2 are now time varying. (*Hint:* Use an example.)
(g) If T_1 and T_2 are nonlinear, then T is nonlinear.
(h) If T_1 and T_2 are stable, then T is stable.
(i) Show by an example that the inverse of parts (c) and (h) do not hold in general.

2.9 Let T be an LTI, relaxed, and BIBO stable system with input $x(n)$ and output $y(n)$. Show that:
(a) If $x(n)$ is periodic with period N [i.e., $x(n) = x(n+N)$ for all $n \geq 0$], the output $y(n)$ tends to a periodic signal with the same period.
(b) If $x(n)$ is bounded and tends to a constant, the output will also tend to a constant.
(c) If $x(n)$ is an energy signal, the output $y(n)$ will also be an energy signal.

2.10 The following input–output pairs have been observed during the operation of a time-invariant system:

$$x_1(n) = \{\underset{\uparrow}{1}, 0, 2\} \xleftrightarrow{T} y_1(n) = \{\underset{\uparrow}{0}, 1, 2\}$$

$$x_2(n) = \{\underset{\uparrow}{0}, 0, 3\} \xleftrightarrow{T} y_2(n) = \{\underset{\uparrow}{0}, 1, 0, 2\}$$

$$x_3(n) = \{\underset{\uparrow}{0}, 0, 0, 1\} \xleftrightarrow{T} y_3(n) = \{1, \underset{\uparrow}{2}, 1\}$$

Can you draw any conclusions regarding the linearity of the system. What is the impulse response of the system?

2.11 The following input–output pairs have been observed during the operation of a linear system:

$$x_1(n) = \{-1, \underset{\uparrow}{2}, 1\} \xleftrightarrow{T} y_1(n) = \{1, 2, \underset{\uparrow}{-1}, 0, 1\}$$

$$x_2(n) = \{1, \underset{\uparrow}{-1}, -1\} \xleftrightarrow{T} y_2(n) = \{-1, \underset{\uparrow}{1}, 0, 2\}$$

$$x_3(n) = \{0, \underset{\uparrow}{1}, 1\} \xleftrightarrow{T} y_3(n) = \{\underset{\uparrow}{1}, 2, 1\}$$

Can you draw any conclusions about the time invariance of this system?

2.12 The only available information about a system consists of N input–output pairs, of signals $y_i(n) = T[x_i(n)]$, $i = 1, 2, \ldots, N$.

(a) What is the class of input signals for which we can determine the output, using the information above, if the system is known to be linear?

(b) The same as above, if the system is known to be time invariant.

2.13 Show that the necessary and sufficient condition for a relaxed LTI system to be BIBO stable is

$$\sum_{n=-\infty}^{\infty} |h(n)| \leq M_h < \infty$$

for some constant M_n.

2.14 Show that:

(a) A relaxed linear system is causal if and only if for any input $x(n)$ such that

$$x(n) = 0 \text{ for } n < n_0 \Rightarrow y(n) = 0 \qquad \text{for } n < n_0$$

(b) A relaxed LTI system is causal if and only if

$$h(n) = 0 \qquad \text{for } n < 0$$

2.15 (a) Show that for any real or complex constant a, and any finite integer numbers M and N, we have

$$\sum_{n=M}^{N} a^n = \begin{cases} \dfrac{a^M - a^{N+1}}{1-a}, & \text{if } a \neq 1 \\ N - M + 1, & \text{if } a = 1 \end{cases}$$

(b) Show that if $|a| < 1$, then

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$$

2.16 (a) If $y(n) = x(n) * h(n)$, show that $\sum_y = \sum_x \sum_h$, where $\sum_x = \sum_{n=-\infty}^{\infty} x(n)$.

(b) Compute the convolution $y(n) = x(n) * h(n)$ of the following signals and check the correctness of the results by using the test in (a).

(1) $x(n) = \{1, 2, 4\}, h(n) = \{1, 1, 1, 1, 1\}$

(2) $x(n) = \{1, 2, -1\}, h(n) = x(n)$

(3) $x(n) = \{0, 1, -2, 3, -4\}, h(n) = \{\frac{1}{2}, \frac{1}{2}, 1, \frac{1}{2}\}$

(4) $x(n) = \{1, 2, 3, 4, 5\}, h(n) = \{1\}$

(5) $x(n) = \{\underset{\uparrow}{1}, -2, 3\}, h(n) = \{\underset{\uparrow}{0}, 0, 1, 1, 1, 1\}$

(6) $x(n) = \{\underset{\uparrow}{0}, 0, 1, 1, 1, 1\}, h(n) = \{1, \underset{\uparrow}{-2}, 3\}$

(7) $x(n) = \{\underset{\uparrow}{0}, 1, 4, -3\}, h(n) = \{\underset{\uparrow}{1}, 0, -1, -1\}$

(8) $x(n) = \{\underset{\uparrow}{1}, 1, 2\}, h(n) = u(n)$

(9) $x(n) = \{1, 1, \underset{\uparrow}{0}, 1, 1\}, h(n) = \{1, -2, -3, \underset{\uparrow}{4}\}$

(10) $x(n) = \{1, 2, \underset{\uparrow}{0}, 2, 1\} h(n) = x(n)$

(11) $x(n) = (\frac{1}{2})^n u(n), h(n) = (\frac{1}{4})^n u(n)$

2.17 Compute and plot the convolutions $x(n) * h(n)$ and $h(n) * x(n)$ for the pairs of signals shown in Fig. P2.17.

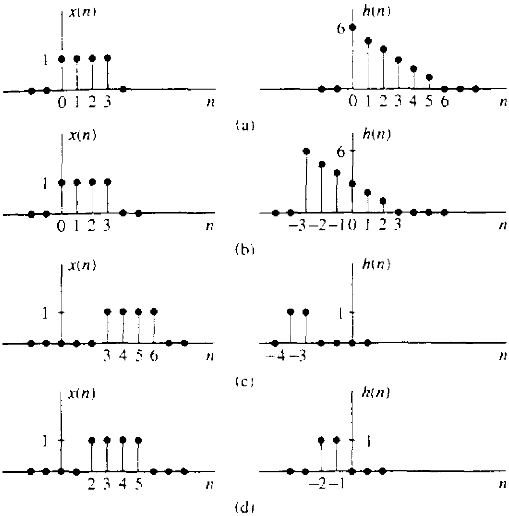


Figure P2.17

2.18 Determine and sketch the convolution $y(n)$ of the signals

$$x(n) = \begin{cases} \frac{1}{3}n, & 0 \le n \le 6 \\ 0, & \text{elsewhere} \end{cases}$$

$$h(n) = \begin{cases} 1, & -2 \le n \le 2 \\ 0, & \text{elsewhere} \end{cases}$$

(a) Graphically
(b) Analytically

2.19 Compute the convolution $y(n)$ of the signals

$$x(n) = \begin{cases} \alpha^n, & -3 \le n \le 5 \\ 0, & \text{elsewhere} \end{cases}$$

$$h(n) = \begin{cases} 1, & 0 \le n \le 4 \\ 0, & \text{elsewhere} \end{cases}$$

2.20 Consider the following three operations.
(a) Multiply the integer numbers: 131 and 122.
(b) Compute the convolution of signals: $\{1, 3, 1\} * \{1, 2, 2\}$.
(c) Multiply the polynomials: $1 + 3z + z^2$ and $1 + 2z + 2z^2$.
(d) Repeat part (a) for the numbers 1.31 and 12.2.
(e) Comment on your results.

2.21 Compute the convolution $y(n) = x(n) * h(n)$ of the following pairs of signals.
(a) $x(n) = a^n u(n)$, $h(n) = b^n u(n)$ when $a \ne b$ and when $a = b$

(b) $x(n) = \begin{cases} 1, & n = -2, 0, 1 \\ 2, & n = -1 \\ 0, & \text{elsewhere} \end{cases}$

$h(n) = \delta(n) - \delta(n-1) + \delta(n-4) + \delta(n-5)$

(c) $x(n) = u(n+1) - u(n-4) - \delta(n-5)$
$h(n) = [u(n+2) - u(n-3)] \cdot (3 - |n|)$

(d) $x(n) = u(n) - u(n-5)$
$h(n) = u(n-2) - u(n-8) + u(n-11) - u(n-17)$

2.22 Let $x(n)$ be the input signal to a discrete-time filter with impulse response $h_i(n)$ and let $y_i(n)$ be the corresponding output.

(a) Compute and sketch $x(n)$ and $y_i(n)$ in the following cases, using the same scale in all figures.

$$\begin{aligned}
x(n) &= \{1, 4, 2, 3, 5, 3, 3, 4, 5, 7, 6, 9\} \\
h_1(n) &= \{1, 1\} \\
h_2(n) &= \{1, 2, 1\} \\
h_3(n) &= \{\tfrac{1}{2}, \tfrac{1}{2}\} \\
h_4(n) &= \{\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}\} \\
h_5(n) &= \{\tfrac{1}{4}, -\tfrac{1}{2}, \tfrac{1}{4}\}
\end{aligned}$$

Sketch $x(n)$, $y_1(n)$, $y_2(n)$ on one graph and $x(n)$, $y_3(n)$, $y_4(n)$, $y_5(n)$ on another graph

(b) What is the difference between $y_1(n)$ and $y_2(n)$, and between $y_3(n)$ and $y_4(n)$?

(c) Comment on the smoothness of $y_2(n)$ and $y_4(n)$. Which factors affect the smoothness?

(d) Compare $y_4(n)$ with $y_5(n)$. What is the difference? Can you explain it?

(e) Let $h_6(n) = \{\frac{1}{2}, -\frac{1}{2}\}$. Compute $y_6(n)$. Sketch $x(n)$, $y_2(n)$, and $y_6(n)$ on the same figure and comment on the results.

2.23 The discrete-time system

$$y(n) = ny(n-1) + x(n) \qquad n \geq 0$$

is at rest [i.e., $y(-1) = 0$]. Check if the system is linear time invariant and BIBO stable.

2.24 Consider the signal $\gamma(n) = a^n u(n)$, $0 < a < 1$.

(a) Show that any sequence $x(n)$ can be decomposed as

$$x(n) = \sum_{n=-\infty}^{\infty} c_k \gamma(n-k)$$

and express c_k in terms of $x(n)$.

(b) Use the properties of linearity and time invariance to express the output $y(n) = \mathcal{T}[x(n)]$ in terms of the input $x(n)$ and the signal $g(n) = \mathcal{T}[\gamma(n)]$, where $\mathcal{T}[\cdot]$ is an LTI system.

(c) Express the impulse response $h(n) = \mathcal{T}[\delta(n)]$ in terms of $g(n)$.

2.25 Determine the zero-input response of the system described by the second-order difference equation

$$x(n) - 3y(n-1) - 4y(n-2) = 0$$

2.26 Determine the particular solution of the difference equation

$$y(n) = \tfrac{5}{6}y(n-1) - \tfrac{1}{6}y(n-2) + x(n)$$

when the forcing function is $x(n) = 2^n u(n)$.

2.27 Determine the response $y(n)$, $n \geq 0$, of the system described by the second-order difference equation

$$y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1)$$

to the input $x(n) = 4^n u(n)$.

2.28 Determine the impulse response of the following causal system:

$$y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1)$$

2.29 Let $x(n)$, $N_1 \leq n \leq N_2$ and $h(n)$, $M_1 \leq n \leq M_2$ be two finite-duration signals.

(a) Determine the range $L_1 \leq n \leq L_2$ of their convolution, in terms of N_1, N_2, M_1 and M_2.

(b) Determine the limits of the cases of partial overlap from the left, full overlap, and partial overlap from the right. For convenience, assume that $h(n)$ has shorter duration than $x(n)$.

(c) Illustrate the validity of your results by computing the convolution of the signals

$$x(n) = \begin{cases} 1, & -2 \leq n \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$

$$h(n) = \begin{cases} 2, & -1 \leq n \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

2.30 Determine the impulse response and the unit step response of the systems described by the difference equation

(a) $y(n) = 0.6y(n-1) - 0.08y(n-2) + x(n)$

(b) $y(n) = 0.7y(n-1) - 0.1y(n-2) + 2x(n) - x(n-2)$

2.31 Consider a system with impulse response

$$h(n) = \begin{cases} (\frac{1}{2})^n, & 0 \leq n \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$

Determine the input $x(n)$ for $0 \leq n \leq 8$ that will generate the output sequence

$$y(n) = \{\underset{\uparrow}{1}, 2, 2.5, 3, 3, 3, 2, 1, 0, \ldots\}$$

2.32 Consider the interconnection of LTI systems as shown in Fig. P2.32.

(a) Express the overall impulse response in terms of $h_1(n)$, $h_2(n)$, $h_3(n)$, and $h_4(n)$.

(b) Determine $h(n)$ when

$$h_1(n) = \{\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{2}\}$$

$$h_2(n) = h_3(n) = (n+1)u(n)$$

$$h_4(n) = \delta(n-2)$$

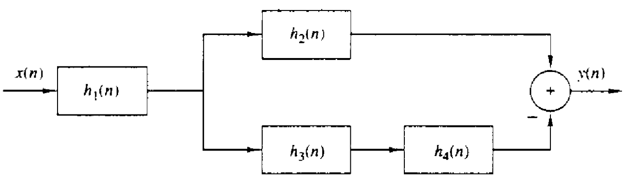


Figure P2.32

(c) Determine the response of the system in part (b) if

$$x(n) = \delta(n+2) + 3\delta(n-1) - 4\delta(n-3)$$

2.33 Consider the system in Fig. P2.33 with $h(n) = a^n u(n)$, $-1 < a < 1$. Determine the response $y(n)$ of the system to the excitation

$$x(n) = u(n+5) - u(n-10)$$

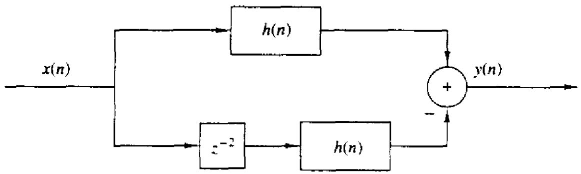


Figure P2.33

2.34 Compute and sketch the step response of the system

$$y(n) = \frac{1}{M} \sum_{k=0}^{M-1} x(n-k)$$

2.35 Determine the range of values of the parameter a for which the linear time-invariant system with impulse response

$$h(n) = \begin{cases} a^n, & n \geq 0, n \text{ even} \\ 0, & \text{otherwise} \end{cases}$$

is stable.

2.36 Determine the response of the system with impulse response

$$h(n) = a^n u(n)$$

to the input signal

$$x(n) = u(n) - u(n-10)$$

(*Hint:* The solution can be obtained easily and quickly by applying the linearity and time-invariance properties to the result in Example 2.3.5.)

2.37 Determine the response of the (relaxed) system characterized by the impulse response

$$h(n) = (\tfrac{1}{2})^n u(n)$$

to the input signal

$$x(n) = \begin{cases} 1, & 0 \leq n < 10 \\ 0, & \text{otherwise} \end{cases}$$

2.38 Determine the response of the (relaxed) system characterized by the impulse response

$$h(n) = (\tfrac{1}{2})^n u(n)$$

to the input signals

(a) $x(n) = 2^n u(n)$

(b) $x(n) = u(-n)$

2.39 Three systems with impulse responses $h_1(n) = \delta(n) - \delta(n-1)$, $h_2(n) = h(n)$, and $h_3(n) = u(n)$, are connected in cascade.

(a) What is the impulse response, $h_c(n)$, of the overall system?

(b) Does the order of the interconnection affect the overall system?

2.40 (a) Prove and explain graphically the difference between the relations

$$x(n)\delta(n-n_0) = x(n_0)\delta(n-n_0) \qquad \text{and} \qquad x(n) * \delta(n-n_0) = x(n-n_0)$$

(b) Show that a discrete-time system, which is described by a convolution summation, is LTI and relaxed,

(c) What is the impulse response of the system described by $y(n) = x(n-n_0)$?

2.41 Two signals $s(n)$ and $v(n)$ are related through the following difference equations

$$s(n) + a_1 s(n-1) + \cdots + a_N s(n-N) = b_0 v(n)$$

Design the block diagram realization of:

(a) The system that generates $s(n)$ when excited by $v(n)$.

(b) The system that generates $v(n)$ when excited by $s(n)$.

(c) What is the impulse response of the cascade interconnection of systems in parts (a) and (b)?

2.42 Compute the zero-state response of the system described by the difference equation

$$y(n) + \tfrac{1}{2}y(n-1) = x(n) + 2x(n-2)$$

to the input

$$x(n) = \{1, 2, \underset{\uparrow}{3}, 4, 2, 1\}$$

by solving the difference equation recursively.

2.43 Determine the direct form II realization for each of the following LTI systems.

(a) $2y(n) + y(n-1) - 4y(n-3) = x(n) + 3x(n-5)$

(b) $y(n) = x(n) - x(n-1) + 2x(n-2) - 3x(n-4)$

2.44 Consider the discrete-time system shown in Fig. P2.44.

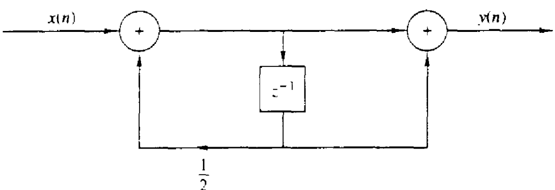


Figure P2.44

(a) Compute the 10 first samples of its impulse response.

(b) Find the input–output relation.

(c) Apply the input $x(n) = \{\underset{\uparrow}{1}, 1, 1, \ldots\}$ and compute the first 10 samples of the output.

(d) Compute the first 10 samples of the output for the input given in part (c) by using convolution.

(e) Is the system causal? Is it stable?

2.45 Consider the system described by the difference equation

$$y(n) = ay(n-1) + bx(n)$$

(a) Determine b in terms of a so that

$$\sum_{n=-\infty}^{\infty} h(n) = 1$$

(b) Compute the zero-state step response $s(n)$ of the system and choose b so that $s(\infty) = 1$.

(c) Compare the values of b obtained in parts (a) and (b). What did you notice?

2.46 A discrete-time system is realized by the structure shown in Fig. P2.46.

(a) Determine the impulse response.

(b) Determine a realization for its inverse system, that is, the system which produces $x(n)$ as an output when $y(n)$ is used as an input.

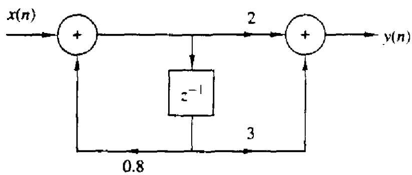


Figure P2.46

2.47 Consider the discrete-time system shown in Fig. P2.47.

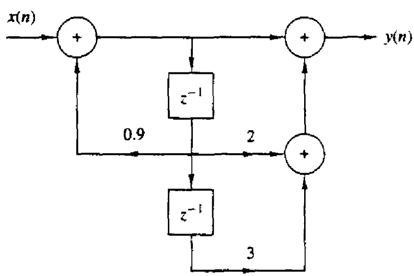


Figure P2.47

(a) Compute the first six values of the impulse response of the system.

(b) Compute the first six values of the zero-state step response of the system.

(c) Determine an analytical expression for the impulse response of the system.

2.48 Determine and sketch the impulse response of the following systems for $n = 0, 1, \ldots, 9$.

(a) Fig. P2.48(a).

(b) Fig. P2.48(b).

(c) Fig. P2.48(c).

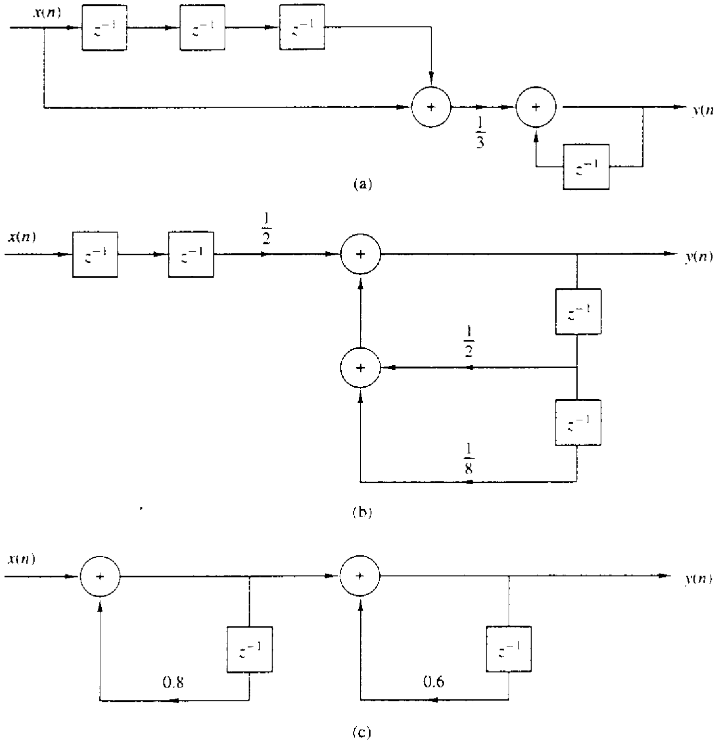


Figure P2.48

(d) Classify the systems above as FIR or IIR.

(e) Find an explicit expression for the impulse response of the system in part (c).

2.49 Consider the systems shown in Fig. P2.49.

(a) Determine and sketch their impulse responses $h_1(n)$, $h_2(n)$, and $h_3(n)$.

(b) Is it possible to choose the coefficients of these systems in such a way that

$$h_1(n) = h_2(n) = h_3(n)$$

2.50 Consider the system shown in Fig. P2.50.

(a) Determine its impulse response $h(n)$.

(b) Show that $h(n)$ is equal to the convolution of the following signals.

$$h_1(n) = \delta(n) + \delta(n-1)$$

$$h_2(n) = (\tfrac{1}{2})^n u(n)$$

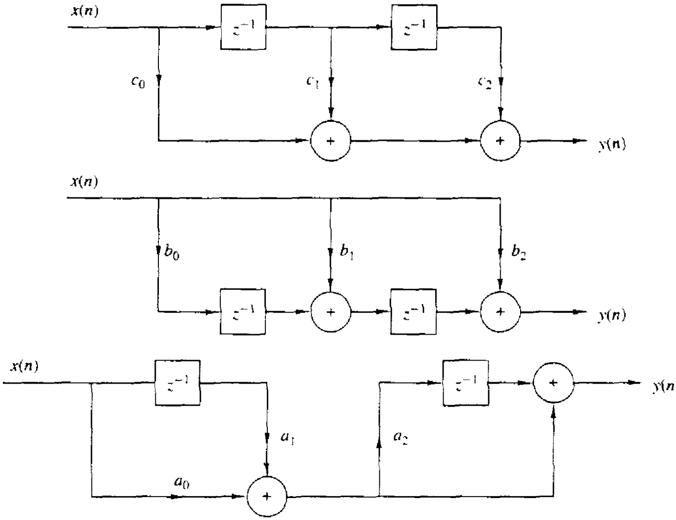


Figure P2.49

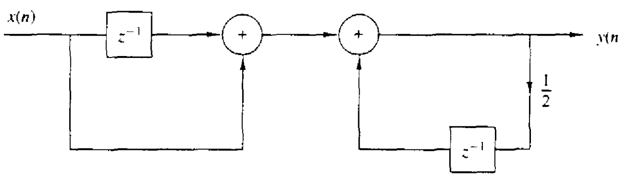


Figure P2.50

2.51 Compute the sketch the convolution $y_i(n)$ and correlation $r_i(n)$ sequences for the following pair of signals and comment on the results obtained.

(a) $x_1(n) = \{\underset{\uparrow}{1}, 2, 4\}$ $\quad h_1(n) = \{\underset{\uparrow}{1}, 1, 1, 1, 1\}$

(b) $x_2(n) = \{\underset{\uparrow}{0}, 1, -2, 3, -4\}$ $\quad h_2(n) = \{\frac{1}{2}, 1, \underset{\uparrow}{2}, 1, \frac{1}{2}\}$

(c) $x_3(n) = \{\underset{\uparrow}{1}, 2, 3, 4\}$ $\quad h_3(n) = \{\underset{\uparrow}{4}, 3, 2, 1\}$

(d) $x_4(n) = \{\underset{\uparrow}{1}, 2, 3, 4\}$ $\quad h_4(n) = \{\underset{\uparrow}{1}, 2, 3, 4\}$

2.52 The zero-state response of a causal LTI system to the input $x(n) = \{\underset{\uparrow}{1}, 3, 3, 1\}$ is $y(n) = \{\underset{\uparrow}{1}, 4, 6, 4, 1\}$. Determine its impulse response.

2.53 Prove by direct substitution the equivalence of equations (2.5.9) and (2.5.10), which describe the direct form II structure, to the relation (2.5.6), which describes the direct form I structure.

2.54 Determine the response $y(n)$, $n \geq 0$ of the system described by the second-order difference equation

$$y(n) - 4y(n-1) + 4y(n-2) = x(n) - x(n-1)$$

when the input is

$$x(n) = (-1)^n u(n)$$

and the initial conditions are $y(-1) = y(-2) = 0$.

2.55 Determine the impulse response $h(n)$ for the system described by the second-order difference equation

$$y(n) - 4y(n-1) + 4y(n-2) = x(n) - x(n-1)$$

2.56 Show that any discrete-time signal $x(n)$ can be expressed as

$$x(n) = \sum_{k=-\infty}^{\infty} [x(k) - x(k-1)]u(n-k)$$

where $u(n-k)$ is a unit step delayed by k units in time, that is,

$$u(n-k) = \begin{cases} 1, & n \geq k \\ 0, & \text{otherwise} \end{cases}$$

2.57 Show that the output of an LTI system can be expressed in terms of its unit step response $s(n)$ as follows.

$$y(n) = \sum_{k=-\infty}^{\infty} [s(k) - s(k-1)]x(n-k)$$

$$= \sum_{k=-\infty}^{\infty} [x(k) - x(k-1)]s(n-k)$$

2.58 Compute the correlation sequences $r_{xx}(l)$ and $r_{xy}(l)$ for the following signal sequences.

$$x(n) = \begin{cases} 1, & n_0 - N \leq n \leq n_0 + N \\ 0, & \text{otherwise} \end{cases}$$

$$y(n) = \begin{cases} 1, & -N \leq n \leq N \\ 0, & \text{otherwise} \end{cases}$$

2.59 Determine the autocorrelation sequences of the following signals.

(a) $x(n) = \{1, \underset{\uparrow}{2}, 1, 1\}$

(b) $y(n) = \{1, \underset{\uparrow}{1}, 2, 1\}$

What is your conclusion?

2.60 What is the normalized autocorrelation sequence of the signal $x(n)$ given by

$$x(n) = \begin{cases} 1, & -N \leq n \leq N \\ 0, & \text{otherwise} \end{cases}$$

2.61 An audio signal $s(t)$ generated by a loudspeaker is reflected at two different walls with reflection coefficients r_1 and r_2. The signal $x(t)$ recorded by a microphone close to the loudspeaker, after sampling, is

$$x(n) = s(n) + r_1 s(n - k_1) + r_2 s(n - k_2)$$

where k_1 and k_2 are the delays of the two echoes.

(a) Determine the autocorrelation $r_{xx}(l)$ of the signal $x(n)$.

(b) Can we obtain r_1, r_2, k_1, and k_2 by observing $r_{xx}(l)$?

(c) What happens if $r_2 = 0$?

2.62* *Time-delay estimation in radar* Let $x_a(t)$ be the transmitted signal and $y_a(t)$ be the received signal in a radar system, where

$$y_a(t) = a x_a(t - t_d) + v_a(t)$$

and $v_a(t)$ is additive random noise. The signals $x_a(t)$ and $y_a(t)$ are sampled in the receiver, according to the sampling theorem, and are processed digitally to determine the time delay and hence the distance of the object. The resulting discrete-time signals are

$$\begin{aligned} x(n) &= x_a(nT) \\ y(n) &= y_a(nT) = a x_a(nT - DT) + v_a(nT) \\ &\stackrel{\Delta}{=} a x(n - D) + v(n) \end{aligned}$$

(a) Explain how we can measure the delay D by computing the crosscorrelation $r_{xy}(l)$.

(b) Let $x(n)$ be the 13-point *Barker sequence*

$$x(n) = \{+1, +1, +1, +1, +1, -1, -1, +1, +1, -1, +1, -1, +1\}$$

and $v(n)$ be a Gaussian random sequence with zero mean and variance $\sigma^2 = 0.01$. Write a program that generates the sequence $y(n)$, $0 \leq n \leq 199$ for $a = 0.9$ and $D = 20$. Plot the signals $x(n)$, $y(n)$, $0 \leq n \leq 199$.

(c) Compute and plot the crosscorrelation $r_{xy}(l)$, $0 \leq l \leq 59$. Use the plot to estimate the value of the delay D.

(d) Repeat parts (b) and (c) for $\sigma^2 = 0.1$ and $\sigma^2 = 1$.

(e) Repeat parts (b) and (c) for the signal sequence

$$\begin{aligned} x(n) = \{&-1, -1, -1, +1, +1, +1, +1, -1, \\ &+1, -1, +1, +1, -1, -1, +1\} \end{aligned}$$

which is obtained from the four-stage feedback shift register shown in Fig. P2.62.

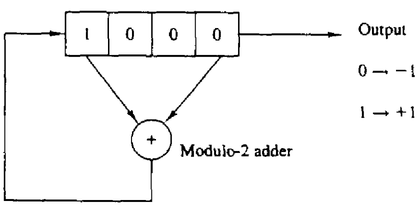


Figure P2.61 Linear feedback shift register.

Note that $x(n)$ is just one period of the periodic sequence obtained from the feedback shift register.

(f) Repeat parts (b) and (c) for a sequence of period $N = 2^7 - 1$, which is obtained from a seven-stage feedback shift register. Table 2.3 gives the stages connected to the modulo-2 adder for (maximal-length) shift-register sequences of length $N = 2^m - 1$.

TABLE 2.3 SHIFT-REGISTER CONNECTIONS FOR GENERATING MAXIMAL-LENGTH SEQUENCES

m	Stages Connected to Modulo-2 Adder
1	1
2	1, 2
3	1, 3
4	1, 4
5	1, 4
6	1, 6
7	1, 7
8	1, 5, 6, 7
9	1, 6
10	1, 8
11	1, 10
12	1, 7, 9, 12
13	1, 10, 11, 13
14	1, 5, 9, 14
15	1, 15
16	1, 5, 14, 16
17	1, 15

2.63* *Implementation of LTI systems* Consider the recursive discrete-time system described by the difference equation

$$y(n) = -a_1 y(n-1) - a_2 y(n-2) + b_0 x(n)$$

where $a_1 = -0.8$, $a_2 = 0.64$, and $b_0 = 0.866$.

(a) Write a program to compute and plot the impulse response $h(n)$ of the system for $0 \le n \le 49$.

(b) Write a program to compute and plot the zero-state step response $s(n)$ of the system for $0 \le n \le 100$.

(c) Define an FIR system with impulse response $h_{FIR}(n)$ given by

$$h_{FIR}(n) = \begin{cases} h(n), & 0 \le n \le 19 \\ 0, & \text{elsewhere} \end{cases}$$

where $h(n)$ is the impulse response computed in part (a). Write a program to compute and plot its step response.

(d) Compare the results obtained in parts (b) and (c) and explain their similarities and differences.

2.64* Write a computer program that computes the overall impulse response $h(n)$ of the system shown in Fig. P2.64 for $0 \le n \le 99$. The systems $\mathcal{T}_1$, $\mathcal{T}_2$, $\mathcal{T}_3$, and $\mathcal{T}_4$ are specified by

$$\mathcal{T}_1 : h_1(n) = \{\underset{\uparrow}{1}, \tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{8}, \tfrac{1}{16}, \tfrac{1}{32}\}$$

$$\mathcal{T}_2 : h_2(n) = \{\underset{\uparrow}{1}, 1, 1, 1, 1\}$$

$$\mathcal{T}_3 : y_3(n) = \tfrac{1}{4}x(n) + \tfrac{1}{2}x(n-1) + \tfrac{1}{4}x(n-2)$$

$$\mathcal{T}_4 : y(n) = 0.9y(n-1) - 0.81y(n-2) + v(n) + v(n-1)$$

Plot $h(n)$ for $0 \le n \le 99$.

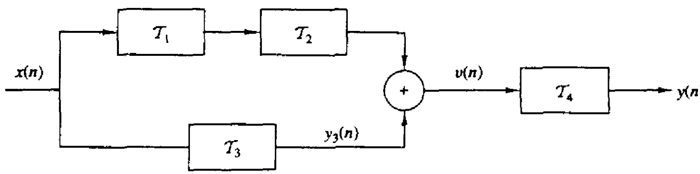


Figure P2.64

3

The Z-Transform and Its Application to the Analysis of LTI Systems

Transform techniques are an important tool in the analysis of signals and linear time-invariant (LTI) systems. In this chapter we introduce the z-transform, develop its properties, and demonstrate its importance in the analysis and characterization of linear time-invariant systems.

The z-transform plays the same role in the analysis of discrete-time signals and LTI systems as the Laplace transform does in the analysis of continuous-time signals and LTI systems. For example, we shall see that in the z-domain (complex z-plane) the convolution of two time-domain signals is equivalent to multiplication of their corresponding z-transforms. This property greatly simplifies the analysis of the response of an LTI system to various signals. In addition, the z-transform provides us with a means of characterizing an LTI system, and its response to various signals, by its pole–zero locations.

We begin this chapter by defining the z-transform. Its important properties are presented in Section 3.2. In Section 3.3 the transform is used to characterize signals in terms of their pole–zero patterns. Section 3.4 describes methods for inverting the z-transform of a signal so as to obtain the time-domain representation of the signal. The one-sided z-transform is treated in Section 3.5 and used to solve linear difference equations with nonzero initial conditions. The chapter concludes with a discussion on the use of the z-transform in the analysis of LTI systems.

3.1 THE *Z*-TRANSFORM

In this section we introduce the z-transform of a discrete-time signal, investigate its convergence properties, and briefly discuss the inverse z-transform.

3.1.1 The Direct z-Transform

The z-transform of a discrete-time signal $x(n)$ is defined as the power series

$$X(z) \equiv \sum_{n=-\infty}^{\infty} x(n)z^{-n} \tag{3.1.1}$$

where z is a complex variable. The relation (3.1.1) is sometimes called the *direct z-transform* because it transforms the time-domain signal $x(n)$ into its complex-plane representation $X(z)$. The inverse procedure [i.e., obtaining $x(n)$ from $X(z)$] is called the *inverse z-transform* and is examined briefly in Section 3.1.2 and in more detail in Section 3.4.

For convenience, the z-transform of a signal $x(n)$ is denoted by

$$X(z) \equiv Z\{x(n)\} \tag{3.1.2}$$

whereas the relationship between $x(n)$ and $X(z)$ is indicated by

$$x(n) \overset{z}{\longleftrightarrow} X(z) \tag{3.1.3}$$

Since the z-transform is an infinite power series, it exists only for those values of z for which this series converges. The *region of convergence* (ROC) of $X(z)$ is the set of all values of z for which $X(z)$ attains a finite value. Thus any time we cite a z-transform we should also indicate its ROC.

We illustrate these concepts by some simple examples.

Example 3.1.1

Determine the z-transforms of the following *finite-duration* signals.

(a) $x_1(n) = \{1, 2, 5, 7, 0, 1\}$

(b) $x_2(n) = \{1, 2, \underset{\uparrow}{5}, 7, 0, 1\}$

(c) $x_3(n) = \{0, 0, 1, 2, 5, 7, 0, 1\}$

(d) $x_4(n) = \{2, 4, \underset{\uparrow}{5}, 7, 0, 1\}$

(e) $x_5(n) = \delta(n)$

(f) $x_6(n) = \delta(n-k), k > 0$

(g) $x_7(n) = \delta(n+k), k > 0$

Solution From definition (3.1.1), we have

(a) $X_1(z) = 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$, ROC: entire z-plane except $z = 0$

(b) $X_2(z) = z^2 + 2z + 5 + 7z^{-1} + z^{-3}$, ROC: entire z-plane except $z = 0$ and $z = \infty$

(c) $X_3(z) = z^{-2} + 2z^{-3} + 5z^{-4} + 7z^{-5} + z^{-7}$, ROC: entire z-plane except $z = 0$

(d) $X_4(z) = 2z^2 + 4z + 5 + 7z^{-1} + z^{-3}$, ROC: entire z-plane except $z = 0$ and $z = \infty$

(e) $X_5(z) = 1$[i.e., $\delta(n) \overset{z}{\longleftrightarrow} 1$], ROC: entire z-plane

(f) $X_6(z) = z^{-k}$[i.e., $\delta(n-k) \overset{z}{\longleftrightarrow} z^{-k}$], $k > 0$, ROC: entire z-plane except $z = 0$

(g) $X_7(z) = z^k$[*i.e.*, $\delta(n+k) \overset{z}{\longleftrightarrow} z^k$], $k > 0$, ROC: entire z-plane except $z = \infty$

From this example it is easily seen that the ROC of a *finite-duration signal* is the entire z-plane, except possibly the points $z = 0$ and/or $z = \infty$. These points are excluded, because $z^k (k > 0)$ becomes unbounded for $z = \infty$ and $z^{-k} (k > 0)$ becomes unbounded for $z = 0$.

From a mathematical point of view the z-transform is simply an alternative representation of a signal. This is nicely illustrated in Example 3.1.1, where we see that the coefficient of z^{-n}, in a given transform, is the value of the signal at time n. In other words, the exponent of z contains the time information we need to identify the samples of the signal.

In many cases we can express the sum of the finite or infinite series for the z-transform in a closed-form expression. In such cases the z-transform offers a compact alternative representation of the signal.

Example 3.1.2

Determine the z-transform of the signal

$$x(n) = (\tfrac{1}{2})^n u(n)$$

Solution The signal $x(n)$ consists of an infinite number of nonzero values

$$x(n) = \{1, (\tfrac{1}{2}), (\tfrac{1}{2})^2, (\tfrac{1}{2})^3, \ldots, (\tfrac{1}{2})^n, \ldots\}$$

The z-transform of $x(n)$ is the infinite power series

$$\begin{aligned} X(z) &= 1 + \tfrac{1}{2}z^{-1} + (\tfrac{1}{2})^2 z^{-2} + (\tfrac{1}{2})^n z^{-n} + \cdots \\ &= \sum_{n=0}^{\infty} (\tfrac{1}{2})^n z^{-n} = \sum_{n=0}^{\infty} (\tfrac{1}{2}z^{-1})^n \end{aligned}$$

This is an infinite geometric series. We recall that

$$1 + A + A^2 + A^3 + \cdots = \frac{1}{1 - A} \qquad \text{if } |A| < 1$$

Consequently, for $|\frac{1}{2}z^{-1}| < 1$, or equivalently, for $|z| > \frac{1}{2}$, $X(z)$ converges to

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \qquad \text{ROC: } |z| > \tfrac{1}{2}$$

We see that in this case, the z-transform provides a compact alternative representation of the signal $x(n)$.

Let us express the complex variable z in polar form as

$$z = re^{j\theta} \tag{3.1.4}$$

where $r = |z|$ and $\theta = \measuredangle z$. Then $X(z)$ can be expressed as

$$X(z)|_{z=re^{j\theta}} = \sum_{n=-\infty}^{\infty} x(n) r^{-n} e^{-j\theta n}$$

In the ROC of $X(z)$, $|X(z)| < \infty$. But

$$\begin{aligned} |X(z)| &= \left| \sum_{n=-\infty}^{\infty} x(n) r^{-n} e^{-j\theta n} \right| \\ &\le \sum_{n=-\infty}^{\infty} |x(n) r^{-n} e^{-j\theta n}| = \sum_{n=-\infty}^{\infty} |x(n) r^{-n}| \end{aligned} \tag{3.1.5}$$

Hence $|X(z)|$ is finite if the sequence $x(n)r^{-n}$ is absolutely summable.

The problem of finding the ROC for $X(z)$ is equivalent to determining the range of values of r for which the sequence $x(n)r^{-n}$ is absolutely summable. To elaborate, let us express (3.1.5) as

$$\begin{aligned} |X(z)| &\le \sum_{n=-\infty}^{-1} |x(n) r^{-n}| + \sum_{n=0}^{\infty} \left| \frac{x(n)}{r^n} \right| \\ &\le \sum_{n=1}^{\infty} |x(-n) r^{n}| + \sum_{n=0}^{\infty} \left| \frac{x(n)}{r^n} \right| \end{aligned} \tag{3.1.6}$$

If $X(z)$ converges in some region of the complex plane, both summations in (3.1.6) must be finite in that region. If the first sum in (3.1.6) converges, there must exist values of r small enough such that the product sequence $x(-n)r^n$, $1 \le n < \infty$, is absolutely summable. Therefore, the ROC for the first sum consists of all points in a circle of some radius r_1, where $r_1 < \infty$, as illustrated in Fig. 3.1a. On the other hand, if the second sum in (3.1.6) converges, there must exist values of r large enough such that the product sequence $x(n)/r^n$, $0 \le n < \infty$, is absolutely summable. Hence the ROC for the second sum in (3.1.6) consists of all points outside a circle of radius $r > r_2$, as illustrated in Fig. 3.1b.

Since the convergence of $X(z)$ requires that both sums in (3.1.6) be finite, it follows that the ROC of $X(z)$ is generally specified as the annular region in the z-plane, $r_2 < r < r_1$, which is the common region where both sums are finite. This region is illustrated in Fig. 3.1c. On the other hand, if $r_2 > r_1$, there is no common region of convergence for the two sums and hence $X(z)$ does not exist.

The following examples illustrate these important concepts.

Example 3.1.3

Determine the z-transform of the signal

$$x(n) = \alpha^n u(n) = \begin{cases} \alpha^n, & n \ge 0 \\ 0, & n < 0 \end{cases}$$

Solution From the definition (3.1.1) we have

$$X(z) = \sum_{n=0}^{\infty} \alpha^n z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n$$

If $|\alpha z^{-1}| < 1$ or equivalently, $|z| > |\alpha|$, this power series converges to $1/(1 - \alpha z^{-1})$.

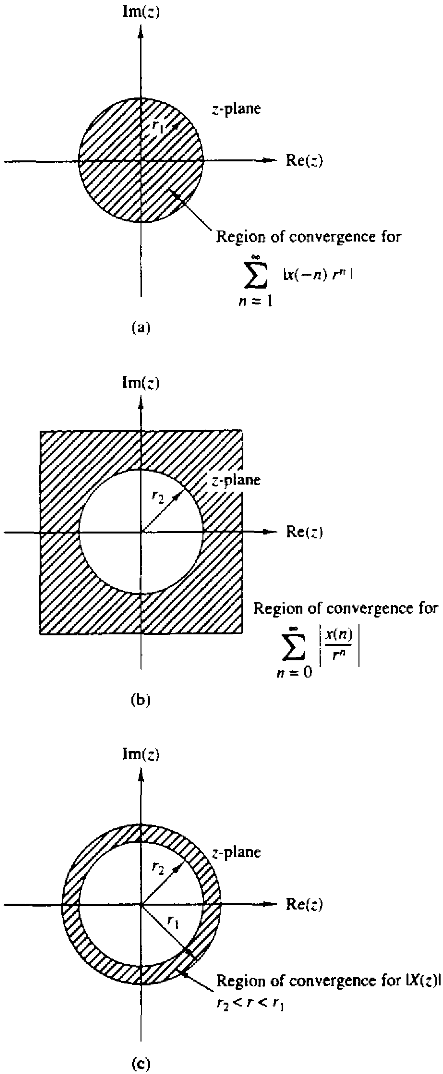


Figure 3.1 Region of convergence for $X(z)$ and its corresponding causal and anticausal components.

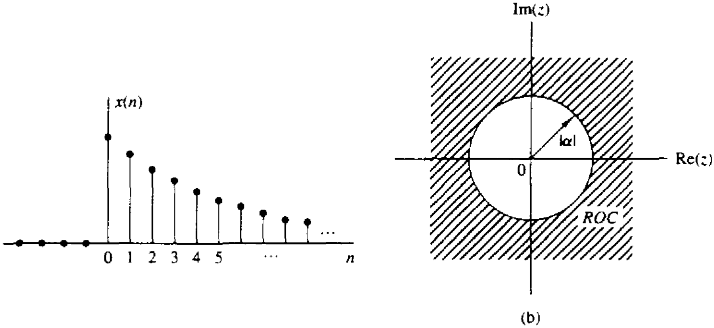


Figure 3.2 The exponential signal $x(n) = \alpha^n u(n)$ (a), and the ROC of its z-transform (b).

Thus we have the z-transform pair

$$x(n) = \alpha^n u(n) \xleftrightarrow{z} X(z) = \frac{1}{1-\alpha z^{-1}} \qquad \text{ROC: } |z| > |\alpha| \tag{3.1.7}$$

The ROC is the exterior of a circle having radius $|\alpha|$. Figure 3.2 shows a graph of the signal $x(n)$ and its corresponding ROC. Note that, in general, α need not be real.

If we set $\alpha = 1$ in (3.1.7), we obtain the z-transform of the unit step signal

$$x(n) = u(n) \xleftrightarrow{z} X(z) = \frac{1}{1-z^{-1}} \qquad \text{ROC: } |z| > 1 \tag{3.1.8}$$

Example 3.1.4

Determine the z-transform of the signal

$$x(n) = -\alpha^n u(-n-1) = \begin{cases} 0, & n \geq 0 \\ -\alpha^n, & n \leq -1 \end{cases}$$

Solution From the definition (3.1.1) we have

$$X(z) = \sum_{n=-\infty}^{-1} (-\alpha^n) z^{-n} = -\sum_{l=1}^{\infty} (\alpha^{-1}z)^l$$

where $l = -n$. Using the formula

$$A + A^2 + A^3 + \cdots = A(1 + A + A^2 + \cdots) = \frac{A}{1-A}$$

when $|A| < 1$ gives

$$X(z) = -\frac{\alpha^{-1}z}{1-\alpha^{-1}z} = \frac{1}{1-\alpha z^{-1}}$$

provided that $|\alpha^{-1}z| < 1$ or, equivalently, $|z| < |\alpha|$. Thus

$$x(n) = -\alpha^n u(-n-1) \xleftrightarrow{z} X(z) = -\frac{1}{1-\alpha z^{-1}} \qquad \text{ROC: } |z| < |\alpha| \tag{3.1.9}$$

The ROC is now the interior of a circle having radius $|\alpha|$. This is shown in Fig. 3.3.

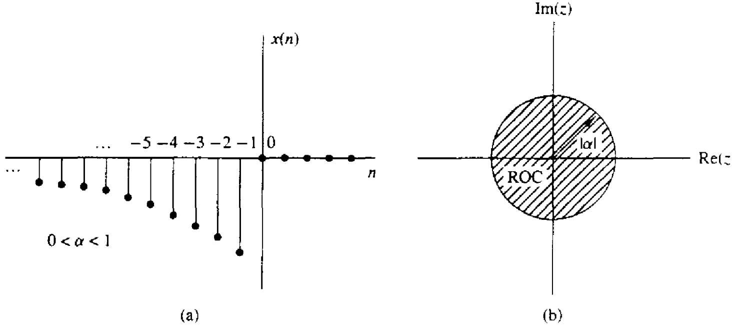


Figure 3.3 Anticausal signal $x(n) = -\alpha^n u(-n-1)$ (a), and the ROC of its z-transform (b).

Examples 3.1.3 and 3.1.4 illustrate two very important issues. The first concerns the uniqueness of the z-transform. From (3.1.7) and (3.1.9) we see that the causal signal $\alpha^n u(n)$ and the anticausal signal $-\alpha^n u(-n-1)$ have identical closed-form expressions for the z-transform, that is,

$$Z\{\alpha^n u(n)\} = Z\{-\alpha^n u(-n-1)\} = \frac{1}{1-\alpha z^{-1}}$$

This implies that a closed-form expression for the z-transform does not uniquely specify the signal in the time domain. The ambiguity can be resolved only if in addition to the closed-form expression, the ROC is specified. In summary, *a discrete-time signal $x(n)$ is uniquely determined by its z-transform $X(z)$ and the region of convergence of $X(z)$*. In this text the term "z-transform" is used to refer to both the closed-form expression and the corresponding ROC. Example 3.1.3 also illustrates the point that *the ROC of a causal signal is the exterior of a circle of some radius r_2 while the ROC of an anticausal signal is the interior of a circle of some radius r_1*. The following example considers a sequence that is nonzero for $-\infty < n < \infty$.

Example 3.1.5

Determine the z-transform of the signal

$$x(n) = \alpha^n u(n) + b^n u(-n-1)$$

Solution From definition (3.1.1) we have

$$X(z) = \sum_{n=0}^{\infty} \alpha^n z^{-n} + \sum_{n=-\infty}^{-1} b^n z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n + \sum_{l=1}^{\infty} (b^{-1} z)^l$$

The first power series converges if $|\alpha z^{-1}| < 1$ or $|z| > |\alpha|$. The second power series converges if $|b^{-1} z| < 1$ or $|z| < |b|$.

In determining the convergence of $X(z)$, we consider two different cases.

Case 1 $|b| < |\alpha|$: In this case the two ROC above do not overlap, as shown in Fig. 3.4(a). Consequently, we cannot find values of z for which both power series converge simultaneously. Clearly, in this case, $X(z)$ does not exist.

Case 2 $|b| > |\alpha|$: In this case there is a ring in the z-plane where both power series converge simultaneously, as shown in Fig. 3.4(b). Then we obtain

$$\begin{aligned} X(z) &= \frac{1}{1-\alpha z^{-1}} - \frac{1}{1-bz^{-1}} \\ &= \frac{b-\alpha}{\alpha + b - z - \alpha b z^{-1}} \end{aligned} \tag{3.1.10}$$

The ROC of $X(z)$ is $|\alpha| < |z| < |b|$.

This example shows that *if there is a ROC for an infinite duration two-sided signal, it is a ring (annular region) in the z-plane.* From Examples 3.1.1, 3.1.3, 3.1.4, and 3.1.5, we see that the ROC of a signal depends on both its duration (finite or infinite) and on whether it is causal, anticausal, or two-sided. These facts are summarized in Table 3.1.

One special case of a two-sided signal is a signal that has infinite duration on the right side but not on the left [i.e., $x(n) = 0$ for $n < n_0 < 0$]. A second case is a signal that has infinite duration on the left side but not on the

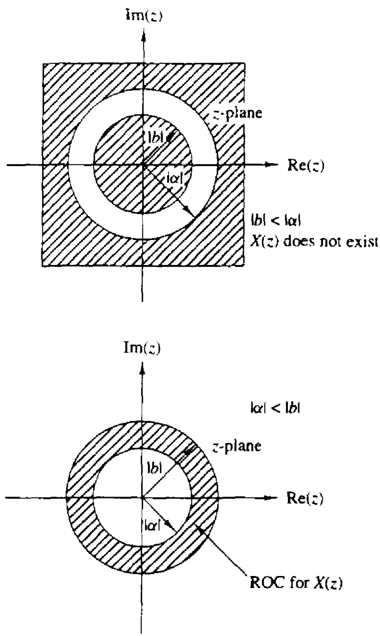
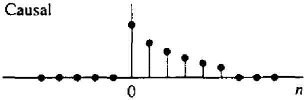
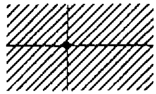
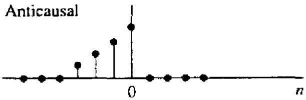
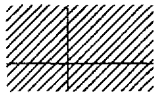
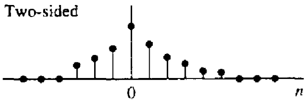
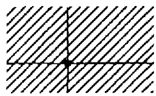
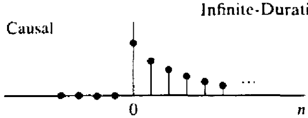
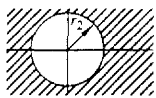
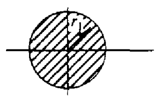
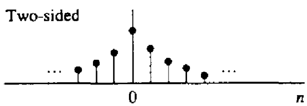
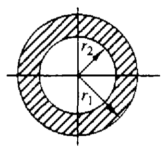


Figure 3.4 ROC for z-transform in Example 3.1.5.

TABLE 3.1 CHARACTERISTIC FAMILIES OF SIGNALS WITH THEIR CORRESPONDING ROC

Signal		ROC	
		Finite-Duration Signals	
Causal			Entire z-plane except $z = 0$
Anticausal			Entire z-plane except $z = \infty$
Two-sided			Entire z-plane except $z = 0$ and $z = \infty$
		Infinite-Duration Signals	
Causal			$\lvert z\rvert > r_2$
Anticausal			$\lvert z\rvert < r_1$
Two-sided			$r_2 < \lvert z\rvert < r_1$

right [i.e., $x(n) = 0$ for $n > n_1 > 0$]. A third special case is a signal that has finite duration on both the left and right sides [i.e., $x(n) = 0$ for $n < n_0 < 0$ and $n > n_1 > 0$]. These types of signals are sometimes called *right-sided, left-sided,* and *finite-duration two-sided,* signals, respectively. The determination of the ROC for these three types of signals is left as an exercise for the reader (Problem 3.5).

Finally, we note that the z-transform defined by (3.1.1) is sometimes referred to as the *two-sided* or *bilateral z-transform*, to distinguish it from the *one-sided* or *unilateral z-transform* given by

$$X^{+}(z) = \sum_{n=0}^{\infty} x(n)z^{-n} \tag{3.1.11}$$

The one-sided z-transform is examined in Section 3.5. In this text we use the expression z-transform exclusively to mean the two-sided z-transform defined by (3.1.1). The term "two-sided" will be used only in cases where we want to resolve any ambiguities. Clearly, if $x(n)$ is causal [i.e., $x(n) = 0$ for $n < 0$], the one-sided and two-sided z-transforms are equivalent. In any other case, they are different.

3.1.2 The Inverse z-Transform

Often, we have the z-transform $X(z)$ of a signal and we must determine the signal sequence. The procedure for transforming from the z-domain to the time domain is called the *inverse z-transform*. An inversion formula for obtaining $x(n)$ from $X(z)$ can be derived by using the *Cauchy integral theorem*, which is an important theorem in the theory of complex variables.

To begin, we have the z-transform defined by (3.1.1) as

$$X(z) = \sum_{k=-\infty}^{\infty} x(k)z^{-k} \tag{3.1.12}$$

Suppose that we multiply both sides of (3.1.12) by z^{n-1} and integrate both sides over a closed contour within the ROC of $X(z)$ which encloses the origin. Such a contour is illustrated in Fig. 3.5. Thus we have

$$\oint_C X(z)z^{n-1}\,dz = \oint_C \sum_{k=-\infty}^{\infty} x(k)z^{n-1-k}\,dz \tag{3.1.13}$$

where C denotes the closed contour in the ROC of $X(z)$, taken in a counterclockwise direction. Since the series converges on this contour, we can interchange the order of integration and summation on the right-hand side of (3.1.13). Thus

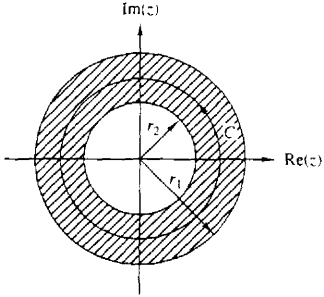


Figure 3.5 Contour C for integral in (3.1.13).

(3.1.13) becomes

$$\oint_C X(z)z^{n-1}dz = \sum_{k=-\infty}^{\infty} x(k)\oint_C z^{n-1-k}dz \tag{3.1.14}$$

Now we can invoke the Cauchy integral theorem, which states that

$$\frac{1}{2\pi j}\oint_C z^{n-1-k}\, dz = \begin{cases} 1, & k = n \\ 0, & k \neq n \end{cases} \tag{3.1.15}$$

where C is any contour that encloses the origin. By applying (3.1.15), the right-hand side of (3.1.14) reduces to $2\pi j x(n)$ and hence the desired inversion formula

$$x(n) = \frac{1}{2\pi j}\oint_C X(z)z^{n-1}\, dz \tag{3.1.16}$$

Although the contour integral in (3.1.16) provides the desired inversion formula for determining the sequence $x(n)$ from the z-transform, we shall not use (3.1.16) directly in our evaluation of inverse z-transforms. In our treatment we deal with signals and systems in the z-domain which have rational z-transforms (i.e., z-transforms that are a ratio of two polynomials). For such z-transforms we develop a simpler method for inversion that stems from (3.1.16) and employs a table lookup.

3.2 PROPERTIES OF THE Z-TRANSFORM

The z-transform is a very powerful tool for the study of discrete-time signals and systems. The power of this transform is a consequence of some very important properties that the transform possesses. In this section we examine some of these properties.

In the treatment that follows, it should be remembered that when we combine several z-transforms, the ROC of the overall transform is, at least, the intersection of the ROC of the individual transforms. This will become more apparent later, when we discuss specific examples.

Linearity. If

$$x_1(n) \overset{z}{\longleftrightarrow} X_1(z)$$

and

$$x_2(n) \overset{z}{\longleftrightarrow} X_2(z)$$

then

$$x(n) = a_1x_1(n) + a_2x_2(n) \overset{z}{\longleftrightarrow} X(z) = a_1X_1(z) + a_2X_2(z) \tag{3.2.1}$$

for any constants a_1 and a_2. The proof of this property follows immediately from the definition of linearity and is left as an exercise for the reader.

The linearity property can easily be generalized for an arbitrary number of signals. Basically, it implies that the z-transform of a linear combination of signals is the same linear combination of their z-transforms. Thus the linearity property helps us to find the z-transform of a signal by expressing the signal as a sum of elementary signals, for each of which, the z-transform is already known.

Example 3.2.1

Determine the z-transform and the ROC of the signal

$$x(n) = [3(2^n) - 4(3^n)]u(n)$$

Solution If we define the signals

$$x_1(n) = 2^n u(n)$$

and

$$x_2(n) = 3^n u(n)$$

then $x(n)$ can be written as

$$x(n) = 3x_1(n) - 4x_2(n)$$

According to (3.2.1), its z-transform is

$$X(z) = 3X_1(z) - 4X_2(z)$$

From (3.1.7) we recall that

$$\alpha^n u(n) \overset{z}{\longleftrightarrow} \frac{1}{1-\alpha z^{-1}} \qquad \text{ROC: } |z| > |\alpha| \tag{3.2.2}$$

By setting $\alpha = 2$ and $\alpha = 3$ in (3.2.2), we obtain

$$x_1(n) = 2^n u(n) \overset{z}{\longleftrightarrow} X_1(z) = \frac{1}{1-2z^{-1}} \qquad \text{ROC: } |z| > 2$$

$$x_2(n) = 3^n u(n) \overset{z}{\longleftrightarrow} X_2(z) = \frac{1}{1-3z^{-1}} \qquad \text{ROC: } |z| > 3$$

The intersection of the ROC of $X_1(z)$ and $X_2(z)$ is $|z| > 3$. Thus the overall transform $X(z)$ is

$$X(z) = \frac{3}{1-2z^{-1}} - \frac{4}{1-3z^{-1}} \qquad \text{ROC: } |z| > 3$$

Example 3.2.2

Determine the z-transform of the signals

(a) $x(n) = (\cos \omega_0 n)u(n)$
(b) $x(n) = (\sin \omega_0 n)u(n)$

Solution

(a) By using Euler's identity, the signal $x(n)$ can be expressed as

$$x(n) = (\cos \omega_0 n)u(n) = \tfrac{1}{2}e^{j\omega_0 n}u(n) + \tfrac{1}{2}e^{-j\omega_0 n}u(n)$$

Thus (3.2.1) implies that

$$X(z) = \tfrac{1}{2}Z\{e^{j\omega_0 n}u(n)\} + \tfrac{1}{2}Z\{e^{-j\omega_0 n}u(n)\}$$

If we set $\alpha = e^{\pm j\omega_0}(|\alpha| = |e^{\pm j\omega_0}| = 1)$ in (3.2.2), we obtain

$$e^{j\omega_0 n}u(n) \stackrel{z}{\longleftrightarrow} \frac{1}{1 - e^{j\omega_0}z^{-1}} \qquad \text{ROC: } |z| > 1$$

and

$$e^{-j\omega_0 n}u(n) \stackrel{z}{\longleftrightarrow} \frac{1}{1 - e^{-j\omega_0}z^{-1}} \qquad \text{ROC: } |z| > 1$$

Thus

$$X(z) = \frac{1}{2}\frac{1}{1 - e^{j\omega_0}z^{-1}} + \frac{1}{2}\frac{1}{1 - e^{-j\omega_0}z^{-1}} \qquad \text{ROC: } |z| > 1$$

After some simple algebraic manipulations we obtain the desired result, namely,

$$(\cos\omega_0 n)u(n) \stackrel{z}{\longleftrightarrow} \frac{1 - z^{-1}\cos\omega_0}{1 - 2z^{-1}\cos\omega_0 + z^{-2}} \qquad \text{ROC: } |z| > 1 \tag{3.2.3}$$

(b) From Euler's identity,

$$x(n) = (\sin\omega_0 n)u(n) = \frac{1}{2j}[e^{j\omega_0 n}u(n) - e^{-j\omega_0 n}u(n)]$$

Thus

$$X(z) = \frac{1}{2j}\left(\frac{1}{1 - e^{j\omega_0}z^{-1}} - \frac{1}{1 - e^{-j\omega_0}z^{-1}}\right) \qquad \text{ROC: } |z| > 1$$

and finally,

$$(\sin\omega_0 n)u(n) \stackrel{z}{\longleftrightarrow} \frac{z^{-1}\sin\omega_0}{1 - 2z^{-1}\cos\omega_0 + z^{-2}} \qquad \text{ROC: } |z| > 1 \tag{3.2.4}$$

Time shifting. If

$$x(n) \stackrel{z}{\longleftrightarrow} X(z)$$

then

$$x(n-k) \stackrel{z}{\longleftrightarrow} z^{-k}X(z) \tag{3.2.5}$$

The ROC of $z^{-k}X(z)$ is the same as that of $X(z)$ except for $z = 0$ if $k > 0$ and $z = \infty$ if $k < 0$. The proof of this property follows immediately from the definition of the z-transform given in (3.1.1)

The properties of linearity and time shifting are the key features that make the z-transform extremely useful for the analysis of discrete-time LTI systems.

Example 3.2.3

By applying the time-shifting property, determine the z-transform of the signals $x_2(n)$ and $x_3(n)$ in Example 3.1.1 from the z-transform of $x_1(n)$.

Solution It can easily be seen that

$$x_2(n) = x_1(n+2)$$